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Experimental and Computational Investigation of Small-Scale Turbopump Performance with a Novel Barske-Type Impeller

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Abstract

Your abstract goes here. Summarise the key motivations, methodology, and findings of your turbopump FYP.

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Nomenclature

Latin symbols

a	Speed of sound (m/s)
A, A_{th}	Area; geometric nozzle throat area (m^2)
b	Turbine rotor blade height (mm)
b_1, b_2	Pump blade axial width at stations 1, 2 (mm)
b_c, h_c	Annular casing width, radial depth (mm)
c	Absolute velocity (m/s)
c_p	Specific heat at constant pressure (J/(kg K))
c_s	Spouting velocity (m/s)
C_H, K_1	Lock/Busemann head-correction constants (−)
$d_0 \dots d_4$	Pump station diameters (mm)
d_m	Turbine mean (pitch) diameter (mm)
f	Friction coefficient (−)
F_R	Radial thrust (N)
g	Gravitational acceleration ($\mathrm{m/s^2}$)
H	Pump head (m); H_{th} theoretical (Euler) head; H_3 head at diffuser throat
h	Specific enthalpy (J/kg); Δh_s isentropic drop
k_d	Barske diffuser kinetic-recovery coefficient (−)
k_N, k_R	Nozzle, rotor velocity coefficients (Weiss) (−)
k_0	Lock shutoff head constant (−)
K_R	Radial thrust coefficient (−)
M	Mach number; M_{w3} rotor-inlet relative Mach (−)
$\dot{m}$	Mass flow rate (kg/s)
n	Shaft speed (rpm)
n_q	Specific speed $n\sqrt{Q}/H^{0.75}$ (rpm, $\mathrm{m^3/s}$, m)
NPSH_a, NPSH_3	Available NPSH; NPSH at 3 % head drop (m)
p	Static pressure (Pa); p_{01}, T_{01} total state at station 1; p_v vapour pressure
P	Power (W); P_{RR} disc friction/churning; P_w ventilation (windage) power
q	Dynamic pressure $\frac{1}{2}\rho c^2$ (Pa)
Q	Volumetric flow rate ($\mathrm{m^3/s}$); Q_L leakage flow
r	Radius (m); recovery factor in eq. (72) only
R	Specific gas constant (J/(kg K))
Re_u	Peripheral Reynolds number $u_2 r_2/\nu_l$ (−)
s_{ax}	Impeller–casing axial clearance (mm)
T	Temperature (K)
u	Blade (peripheral) speed (m/s)
w	Relative velocity (m/s)
Y	Specific work (J/kg)
z	Blade count (−)

Greek symbols

α	Absolute flow angle from circumferential (°)
β	Relative flow / blade angle from circumferential (°)
γ	Heat capacity ratio (−)
ε	Degree of (partial) admission (−)
ζ_D	Diffuser-nozzle loss coefficient (−)
η	Efficiency: pump overall $\rho g Q H/(\tau\omega)$; η_h hydraulic; η_{ts} turbine total-to-static
μ	Dynamic viscosity (always with a state subscript); Mach angle within the MoC set only
ν	Prandtl–Meyer angle (°); kinematic viscosity is written μ/ρ or ν_l
ρ	Density ($\mathrm{kg/m^3}$); ρ_b blade material density
σ	Slip factor (Wiesner), §2 only (−)
τ	Shaft torque (N m)
ϕ	Pump flow coefficient $Q/(\pi d_2 b_2 u_2)$ (−)
ψ	Pump head coefficient $2gH/u_2^2$ (−)
ω	Shaft angular velocity (rad/s); ω_s nondimensional specific speed

Stations (machine-scoped)

Pump	0 suction/eye inlet · 1 blade leading edge · 2 blade tip · 3 diffuser throat · 4 discharge
Turbine	1 stator inlet plenum · 2 nozzle throat · 3 nozzle exit/rotor inlet · 4 rotor exit

Scoped symbol sets (declared here, used only in their subsections)

MoC set (section 2.2.4)	$M^* = c/a^*$, $a^* = \sqrt{\gamma R T^*}$, r^*, R^*, x^*, y^*, μ_k (Mach angle), m_k, $\Delta\nu$
Starting set (section 2.2.4)	M_l^*, M_u^* surface critical Machs; K_{max}^* weight-flow constant; Q vortex parameter; C 2D re
Boundary layer set (Sasman–Cresci)	H_i shape factor, θ momentum thickness, f, C_f, g_w, T_e, $\bar{T}$, s/c

Abbreviations

ADC	Analogue to Digital Converter
BEP	Best Efficiency Point
CFD	Computational Fluid Dynamics
DAQ	Data Acquisition
ESC	Electronic Speed Controller
FEA	Finite Element Analysis
GUI	Graphical User Interface
MoC	Method of Characteristics
P&ID	Piping and Instrumentation Diagram
SLA	Stereolithography (resin printing)

1 Introduction

Rocket engines require high pressures to operate efficiently. Small scale liquid propulsion such as satellite propulsion and amateur liquid roket propulsion runs almost entirely on pressure fed feed systems due to their simplicity and reliability. Larger orbital rockets use turbopumps because the mass of the required tanks becomes prohibative. However, as space launch becomes exponentially more accessible, there is a growing interest in small scale rocket propulsion systems. For example, lunar landers, satellite propulsion and amateur liquid rockets are seeing a resurgence in interest. Due to the historical focus on large turbopumps for such applications, turbopumps at the smaller scale are not as well documented and have different design considerations. A turbopump is a pump and the turbine that drives it, and both are punished by the small scale.

1.1 Small scale rocket pump challenges

Pump driven propulsion is the solution orbital liquid rockets use to solve the tank mass problem. By storing propellents at lower pressures and pumping them to the required pressures allows for a large reduction in tank mass. The pump is usually driven by turbines powered by burning a portion of the propellents, or by the heat generated by the cooling of the engine. In recent years, battery and electric pump technology has advanced to such a point that for smaller rockets such as the Electron rocket by Rocket Lab [?], electric pumps are starting to become a more viable alternative to turbine driven pumps.

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Pumps are generally characterized by their flow rate Q_{opt}, the head H_{opt} and rotor speed n [?]. These three parameters largely decide the pump geometry and type of pump. These three parameters can be related by a quantity known as specific speed, n_q, N_s or ω_s depending on the units and definitions used:

$$n_q = \frac{n\sqrt{Q}}{H^{3/4}} \quad [\text{RPM}, \text{m}^3/\text{s}, \text{m}] \tag{1}$$

Different pump geometries are optimal for different regions of specific speed as can be shown by fig. 1. Most liquid rocket engines operate between a specifc speed n_q of 20 to 40 and utilize the centrifugal radial impeller to maximize efficiency. This works great for large rocket engines, but scaling this down to smaller engines becomes increasingly difficult. All rocket engines require high pressures, but smaller rocket engines require less flow rate. To stay in the specific speed according to eq. (1) that allows for high efficiency in the centrifugal pump requires increasing rotational speed.

Figure 1: Optimum pump geometry and attainable efficiency against specific speed, with the design point of this thesis at $n_q = 6.4$ marked. Most rocket engine pumps sit in the radial impeller optimum of n_q 20 to 40.

To achieve the specific speed of 20 for a smaller rocket engine of around 5 kN requiring 1.0 kg/s of kerosene at 50 bar leads to rotational speeds of around 71 051 rpm. Such high rotational speeds leads to challenging mechanical design conditions. To solve this problem, a single pump can be separated into multiple stages, separating the head rise to increase the specific speed of each stage. This is a common solution to pumping hydrogen. This thesis however addresses the other solution, which is to deal with the lower specific speed.

As shown in fig. 1, at lower specific speeds, the partial emission pump becomes the optimum efficiency pump geometry at specific speeds of 5-10. However, even though these pumps are the most efficient at these low specific speeds, they physically cannot achieve the high efficiencies of a full emission pump. This is the main limitation of trying to operate at low specific speeds.

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Due to its lower efficiency and the lack of pump fed liquid rockets for small engines, there is a lack of literature, design knowledge and experimental data surrounding these pumps. This thesis aims to address this gap by reviewing existing

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literature, designing and manufacturing a partial emission centrifugal pump and comparing its experimental performance specifically in the use case of rocket engines.
The lack of data is specifically a lack of experimental data. Open studies of this regime do exist, but they are numerical. A Barske type impeller at $n_q = 4.7$ was studied with CFD and reached an efficiency of 31.5% [1]. A conventional closed impeller was designed at the same specific speed as this thesis, but it was CFD only, unvalidated, and required closed channels of 0.31 mm width [2]. A partial emission rocket e-pump for a lunar lander was designed for 630 m at 40,000 RPM, again in CFD [3]. What is scarce is open experimental characterisation at this specific speed, with a measured loss breakdown and measured cavitation behaviour, and that is what this work supplies. Balje's selection diagrams show why this corner of the design space is poorly covered, as the efficiency islands collapse below a non-dimensional specific speed of around 0.1, right where this duty point sits [4]. A NASA trade study of low thrust rocket pumps covers the same trade and carries the Barske pump as a candidate [5].

1.2 Small scale rocket turbine challenges

On the other end of the turbopump shaft is the turbine which drives the pump. The turbine faces the same problem as the pump. If using the gas generator cycle, the small turbine leads to lower blade speeds U. To extract the maximum work from from the gas, gas velocity C_0 must be high, leading to a supersonic blade turbine. The high gas velocity relative to blade speed leads to the turbine running at a lower efficiency than the optimum $U/C_0 \approx 0.5$ [**?**]. The smaller mass flow also cannot fill the annulus at a sensible blade height, leading to partial admission where only a fraction of the annulus has gas flowing through it at any given time. This makes the stage lossy and awkward to design. In addition to these design limitations, supersonic turbine design is far less studied than subsonic, where methods are scattered across older sources.
This thesis tackles the design of a turbopump in the context of small scale liquid rocket engines which forces both pump and turbine off optimum design conditions, then demonstrates their coupled performance in a full integration test. The thesis is framed around the design of a 20 000 rpm turbopump using water as a inert simulant for the pumped rocket propellents. Then, a supersonic partial admission impulse turbine is designed to drive the pump using compressed air as the working fluid. The two are then coupled together and tested as a single system to evaluate performance.

1.3 Scope and Limitations

This thesis covers the theory, design, manufacture and testing of a small scale turbopump system on inert propellents. The pump was tested up to 9000 rpm, significantly lower than the design speed of 20 000 rpm in the first and final iterations of the conducted experiments. This means the data validation can only happen for the off-design point and extrapolation to the design point is required. Cryogenic and hot gas design and testing are strictly outside the scope of tis thesis. Operational safety procedures are included in the appendix.

1.4 Aims and Objectives

The aim of this thesis is to investigate the performance of a small scale rocket low specific speed turbopump with a Barske pump and a supersonic partial admission turbine. To this end, the objectives of this thesis are:

1. Pump characterisation: Determine the head-flow curve, efficiency, and cavitation behaviour of a Barske-type impeller across operating speeds
2. Turbine design and model validation: Design a single-stage partial-admission supersonic impulse turbine with a first-principles computational toolchain, and validate those models against the achievable test data.
3. Turbopump integration: Demonstrate coupled turbopump operation at scaled off-design conditions and perform analytical extrapolation to the full 20 000 rpm gas generator regime.
4. Computational toolchain assessment: Assess the 1D/analytical design toolchain against the experimental data, and establish where analytical prediction is sufficient for this class of machine.

1.5 Thesis Structure

Chapter 2 establishes the theoretical background for both machines: specific speed, the Barske impeller and its sizing and loss models, and the supersonic impulse turbine theory including blade profile design, separation and starting. Chapter 3 applies these models to design the actual pump and turbine, including the mechanical and structural design. Chapter 4 describes the experimental setup: the test configurations, the water and air circuits, and the custom instrumentation and data acquisition system. Chapter 5 presents the experimental results and discusses them against the design predictions and prior literature. Chapter 6 draws conclusions against the objectives and recommends future work.

2 Literature and Theory

A note on the age of the methods used in this chapter. Much of the supersonic blading toolchain dates from the 1950s and 60s [6, 7], which can give the impression of outdated technology. This is not the case. The same methods were used in the development of the HM60 flight engine turbine [8], were given a modern doctoral treatment in 2020 [9], and are still the basis of current small turbine design studies [10, 11]. On the pump side, the Barske architecture is seeing renewed interest for electric rocket pumps [3, 1].

2.1 Pump Theory

2.1.1 Losses at Low Specific Speed

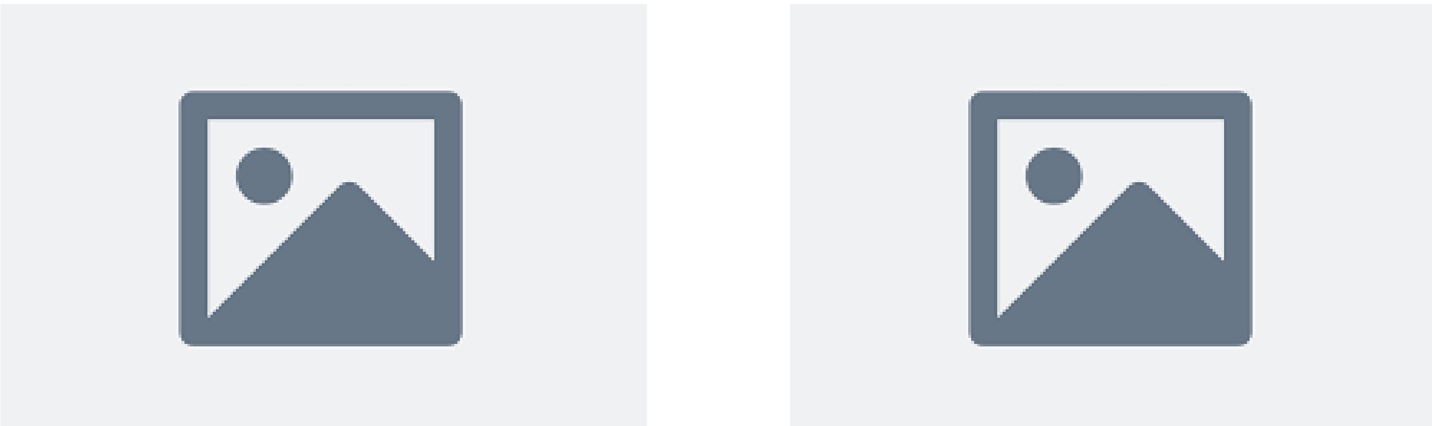
Centrifugal pumpss have multiple main sources of losses. The one that becomes significant at low specific speeds is the disc friction losses. When a rotor object rotates in a fluid, the shear stresses in the fluid cause a drag torque on the object. In most centrifugal pumps the impeller is disc shaped, and the loss is called disc friction loss. . With low specific speed pumps, disk frction is the main source of loss. The disc friction power of a rotating disc scales as

$$P_{RR} = k_{RR}\,\rho\,\omega^3 r_2^5 \tag{2}$$

where k_{RR} is a friction coefficient that falls slowly with Reynolds number [12]. The useful power scales with $\rho g Q H$, so as specific speed falls and the impeller diameter grows relative to the flow, the ratio of disc friction to useful power blows up. At $n_q = 10$, $\psi_{\mathrm{opt}=1}$ and $\mathrm{Re} = 10^7$, the disc friction is approxiomately 30% of the usefulf power of the impeller [**?**]. Gulich suggests that a high head coefficient reduces the diameter and consequently less disk frciont since $P_{RR} \sim D^5$. This is the reason why low specific speed centrifugal pumps physically cannot achieve the same high efficiencies as higher specific speed pumps.

On the other hand, at low specific speeds, impeller losses are low as shown by fig. 3. As a result, surface finish on the impeller blades are largely unimportant, allowing for simpler manufacturing processes. Conversely, collector losses become dominant. Gulich explains [**?**] that for $n_q < 12$ even higher efficiencies are expected than with volutes. This simplifies the design and manufacturing of the pump, which is a significant advantage at small scales where tolerances are both harder to achieve but contribute to more significant losses. This is important as the high fluid velocity at the peripheri makes the surface finish of the annular casing of primary importance for pump efficiency [**?**].

2.1.2 The Barske Partial-Emission Pump



(a) Conventional shrouded full emission impeller

(b) Barske open partial emission impeller

Figure 2: Comparison of the conventional and Barske pump architectures. The Barske impeller is open, has straight radial blades, and discharges into an annular casing with a single tangential diffuser.

Figure 3: Breakdown of pump losses against specific speed, after Gülich. At low specific speed the disc friction dominates while the impeller (blade) losses are small.

The Barske impeller pump design which was developed for the need for manufacturing simplicity and low specific speed applications became one of the highest efficiency designs at this specific speed. The barske impeller operates differently to a normal full emission pump. In a conventional centrifugal pump, impeller blade geometry aims to perfectly guide fluid flow along the blade surfaces in a smooth spiral streamline out of the impeller into the volute [**?**]. The barske impeller is instead designed to induced almost a pure forced vortex flow where $u = \omega r$ and the fluid is essentially rotating as a solid body with the impeller [**?**]. This also explains the unimportance of surface finish on the blades, as low relative velocity between the fluid and the impeller means lower friction losses.

Barske comments that the number of impeller blades does not seem to be critical, with 3-6 being operated successfully. However, altering the number of blades can be used to elminite fluid vibrations [**?**]. Lock provides corrections for blades from 1 to 16 [**?**].

The open impeller choice is to further reduce disc friction losses by minimizing surface area. Conventional impellers uses shrouds to guide the flow and reduce secondary flows. However, as in a Barske pump the impeller serves only to generate the forced vortex flow, this was deemed unnecessary by at least Barske and Lock. This also removes axial thrust loads on the impeller, helping with sealing and mechanical design of the shaft. Also, simplify manufacturing.

Conventional pump design with current design tools produce nonsensical geometries and results at low specific speeds because of one number that is used in the design process, which is the slip coefficient. The slip coefficient determines the angular deflection of the fluid leaving the impeller from the blade angle at the impeller exit, because a finite number of blades cannot perfectly guide the flow. It is given by an empirical regression, the most widely used being Wiesner's fit to Busemann's potential flow solutions [12]:

$$\sigma = 1 - \frac{\sqrt{\sin \beta_{2B}}}{z^{0.7}} \tag{3}$$

where β_{2B} is the blade exit angle from the circumferential direction and z the blade count. The slip factor closes the theoretical head, because the Euler head of eq. (6) with a finite blade count and the outlet velocity triangle becomes

$$\psi_{th} = 2\left(\sigma - \phi_2 \cot \beta_{2B}\right) \tag{4}$$

with ψ and ϕ_2 the head and flow coefficients of eq. (28). The difficulty at low specific speed is now visible in the equation rather than asserted. The duty fixes a high target ψ at a very low ϕ_2, which eq. (4) can only meet by pushing β_{2B} towards the radial and the blade count up, while continuity $\phi_2 = Q/(\pi d_2 b_2 u_2)$ at the same low flow and large diameter simultaneously drives the outlet width b_2 towards zero. The closure does have a root, but it lands on geometry that is either unmanufacturable or outside the validity range of eq. (3), whose own limit on the blade loading is $\varepsilon_{lim} = \exp(-8.16 \sin \beta_{2B}/z)$. This is what motivates the forced vortex Barske architecture, which abandons the attached blade-guided flow model entirely and so removes the slip factor from the design process.

This claim needs to be scoped, because a formally converged conventional design at this specific speed has been published. Kim obtained a converged Wiesner design with $\beta_{2B} = 51.7°$ at the same specific speed as this thesis, but only by mixing correlations from different sources, each used outside its stated validity range, and the resulting geometry required closed channels of $b_2 = 0.31$ mm on a 24 mm impeller [2]. That is exactly the corner of eq. (4) described above: a root exists, but the practical failure of the conventional architecture at this scale is manufacturability and suction robustness rather than pure mathematics. This is explored further with the design attempt in section A and section 5.

2.1.3 Euler and Forced-Vortex Head

Figure 4: Velocity triangle nomenclature and the forced vortex flow field in the Barske impeller, where the fluid rotates as a solid body with $v_\theta = \omega r$.

All theories relating to pump and turbine design can be derived from the conservation of angular momentum by Newton's second law of motion. In turbomachinery, this is usually expressed in terms of the Euler turbine equation:

$$P = T_{shaft}\omega = \rho Q \left(r_2 c_{2u} - r_1 c_{1u}\right)\omega = \rho Q \left(u_2 c_{2u} - u_1 c_{1u}\right) \tag{5}$$

Rewriting this equation in terms of specific work $Y = P/\rho Q$ per unit mass:

$$Y = \left(r_2 c_{2u} - r_1 c_{1u}\right)\omega = u_2 c_{2u} - u_1 c_{1u} \tag{6}$$

derive from bernouli, with rotating frame substitution $c^2 = w^2 - u^2 + 2 \times u \times c_u$ and neglecting compressibility losses:

$$p_1 + \frac{\rho}{2}w_1^2 - \frac{\rho}{2}u_1^2 = p_2 + \frac{\rho}{2}w_2^2 - \frac{\rho}{2}u_2^2$$

and as such the theoretical rise in static pressure:

$$\Delta p_{st} = p_2 - p_1 = \frac{\rho}{2}\left(w_1^2 - w_2^2\right) + \frac{\rho}{2}\left(u_2^2 - u_1^2\right) \tag{7}$$

In Barske, Lobanoff and Lock the relative velocity w is assumed to be negligable, ending up with:

$$\Delta p_{st} = \frac{\rho}{2}\left(u_2^2 - u_1^2\right) \tag{8}$$

Then, the kinetic energy increase assuming negligible inlet prerotation is:

$$\Delta p_{st} = \frac{\rho}{2}u_2^2 \tag{9}$$

eq. (9) is the theoretical total pressure a barske impeller generates at the blade tips. In a barske pump, the diffuser is responsible for converting this kinetic energy into static pressure by slowing down the fluid in an expanding diffuser. Barske introduces a loss coefficient for the kinetic energy which can be recovered through the diffuser to be between 0.2 and 0.5. Barske uses the symbol ψ for this coefficient, but as it conflicts with the commonly used symbol for head coefficient, this paper will instead use c_k. Combining the static and kinetic pressure rise, the total increase in pressure is:

$$p_t = p_{s_{th}} + c_k p_{d_{th}} = \frac{1}{2}\rho\left((1 + c_k)u_2^2 - u_1^2\right) \tag{10}$$

This forms a simple basic model for barske pump performance.

2.1.4 Barske Sizing Method

The barske pump is sized by the methods provided by barske [**?**] with the assitance of Gulich. First the inlet is sized by choosing a reasonable inlet velocity and calculating the inlet area:

$$d_0 = \sqrt{\frac{4\dot{m}}{\pi \rho c_0}} \tag{11}$$

The impeller blade inlet diameter is enlarged by a small amount as some pre-rotation is desirable for increased head. For example, $d_1 \sim 1.1 d_0$.

By substituting $u = \pi n d/60$ into eq. (10), the required impeller tip diameter can be calculated:

$$d_2 = \sqrt{\frac{1}{1+c_k}\left(\frac{7200 p_t}{\rho n^2 \pi^2} + d_1^2\right)} \tag{12}$$

Barske then gives these guidelines for other dimensions of the pump:

$$s_{ax} \leq \min\left(0.75, \frac{d_2}{100}\right) \tag{13}$$

$$b_1 \geq \frac{1}{4} d_1 \tag{14}$$

$$b_2 \geq \max\left(b_1 \frac{d_1}{d_2}, 3 s_{ax}\right) \tag{15}$$

$$\tag{16}$$

For the annular casing sizing, Barske doesn't have very specific numbers, so Gulich's guideline where B/H = 1.5 to 2.5 can be followed:

$$B = 2 s_a x + b_2 \tag{17}$$

$$H = B/2 \tag{18}$$

To size the diffuser, Barske suggests sizing the diffuser throat to be around 0.8 to 0.9 of u_2, with the area of the diffuser outlet to be 3-4 times the area of the diffuser throat.

2.1.5 Efficiency and Disc Friction

Barske assumes that the disc friction losses can be approximated as a circular disc in a rotating fluid. Converting the equation into metric units gives:

$$P_{RR}[\mathrm{W}] = 1956 \times \rho \nu^{0.2} \left(\frac{n}{1000}\right)^{2.8} \left(d_2^{4.6} + 4.6 d_1^{3.6} b_1\right) \tag{19}$$

As the hydraulic efficiency is defined as the useful power out of the pump over the total power into the pump, efficiency can be defined as:

$$\eta_h = \frac{\text{Useful power}}{\text{Input power}} = \frac{P}{P_{th} + P_{RR}} = \frac{Qp}{Qp_{th} + P_{RR}} = \frac{p}{p_{th} + P_{RR}/Q} \tag{20}$$

If total shaft efficiency is desired, the mechanical losses can be included:

$$\eta = \frac{p}{p_{th} + \frac{P_F + P_{Mech}}{Q}} \tag{21}$$

2.1.6 Lock's Refined Model

Lock provides a refined model which takes into account prerotation, nozzle losses and correction coefficients taking into account theoretical two dimensional flow solutions by A. Busemann [?].
By writing the total theoretical pressure gain in head:

$$H_T = \frac{U_2^2 - U_1^2}{2g} + \frac{U_2^2}{2g} \tag{22}$$

Lock uses busemann's corrections and derives a refined total head, taking into account prerotation:

$$H_{TH} = h_0 \frac{U_2^2}{g} - C_H K_1 \left(1 - \frac{Q}{Q_{op}}\right) \frac{D_1}{D_2} \frac{U_2^2}{g} \tag{23}$$

$$H_{SP} = H_{TH} - \underbrace{\zeta_{LD} \left[1 - \left(\frac{D_3}{D_4}\right)^2\right]^2 \frac{V_3^2}{2g}}_{\text{diffuser-nozzle loss}} - \frac{V_4^2}{2g} + \frac{V_0^2}{2g} \tag{24}$$

Where $\zeta_{LD} = 0.194$ is the diffuser loss coefficient from Hydraulic Institute's Pipe Friction Manual [?].
By expressing $V^2/2g$ by $(8/g\pi^2)Q^2/D^4$ with $V = 4Q/\pi D^2$, the head can be expressed as a function of flow rate:

$$H_{SP} = h_0 \frac{U_2^2}{g} - C_H\, K_1 \left(1 - \frac{Q}{Q_{op}}\right) \frac{D_1}{D_2} \frac{U_2^2}{g} - \frac{8Q^2}{g\pi^2} \left(\frac{\zeta_{LD}}{D_3^4} + \frac{1}{D_4^4} - \frac{1}{D_0^4}\right) \tag{25}$$

To find the optimum flow rate used to shape the pump curve, the derivative of the head with respect to flow rate can be taken and set to zero:

$$Q_{op} = \sqrt{\frac{h_0 \frac{U_2^2}{g}}{\frac{24}{g\pi^2}\left(\frac{\zeta_{LD}}{D_3^4}\left(1 - \left(\frac{D_3}{D_4}\right)^2\right) + \frac{1}{D_4^4} - \frac{1}{D_0^4}\right)}} \tag{26}$$

Lock and Barske suggest that the flow cutoff point is limited by the diffuser throat. To find the static pressure at the diffuser throat, reversing the transfer between dynamic head into static head provided by the previous equations :

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$$H_3 = H_{SP} + \frac{8Q^2}{g\pi^2} \left(\frac{\zeta_{LD}}{D_3^4}\left(1 - \left(\frac{D_3}{D_4}\right)^2\right) + \frac{1}{D_4^4} - \frac{1}{D_3^4}\right) \tag{27}$$

Then solving for the Q at which H_3 is equal to vapor pressure head $H_{vap} = \frac{1}{\rho g}\left(p_{vap} - p_{inlet}\right)$ gives the flow cutoff point.
This cutoff mechanism is the key difference between the two pump models used in this thesis. The Euler and Barske models of eq. (9) predict a flat head curve with no flow dependence at all. Lock's model predicts a drooping head curve, and on top of that it predicts that the curve ends at a maximum flow rate where the static head at the diffuser throat reaches vapour pressure and the throat cavitates. Both predictions are directly testable, and the comparison of the measured head curve against both models is one of the main results of section 5. Two limitations that Lock states himself should be kept in mind when his model is compared to data later. He notes that rounding of the blade leading edges reduces the effective head, and that disc friction is treated as a mechanical loss, so it does not affect the head equations and only enters the efficiency [13]. These acknowledged but unmodelled effects matter when the model constants are fitted to data in section 5.

2.1.7 Non-dimensional Performance and Scaling

The measured performance in section 5 is presented in non-dimensional form, so the coefficients are defined here. Following Gülich [12], the head and flow coefficients are

$$\psi = \frac{2gH}{u_2^2}, \qquad \phi = \frac{Q}{\pi d_2 b_2 u_2} \tag{28}$$

The head coefficient measures how much of the natural pressure scale of the rotor, the dynamic head of its tip speed, the machine converts. It is particularly meaningful for this pump: a pure forced vortex delivers $\Delta p = \frac{1}{2}\rho u_2^2$, which is exactly $\psi = 1$, so the measured ψ directly tests whether the forced vortex is achieved. The flow coefficient fixes the shape of the velocity triangles, so two operating points with the same ϕ have the same flow physics regardless of speed.
For a fixed geometry, dimensional analysis of $(Q, H, n, d_2, \rho, \mu)$ reduces the pump behaviour to $\psi = f(\phi, Re)$ with the tip Reynolds number

$$Re_u = \frac{u_2\, r_2}{\nu} \tag{29}$$

Because pump head is Reynolds insensitive to first order, all speeds should collapse onto a single $\psi(\phi)$ curve. Holding ψ and ϕ constant between two speeds gives the affinity laws,

$$Q \propto n, \qquad H \propto n^2, \qquad P \propto n^3 \tag{30}$$

which are the licence for any extrapolation across speed. The losses are not Reynolds insensitive, which is why the efficiency does not collapse the way the head does, and the loss scaling is treated empirically through a loss ratio of the form $(1-\eta) \propto Re^{-a}$.
For cavitation, the suction state is expressed as the net positive suction head,

$$\mathrm{NPSH}_a = \frac{p_{in} - p_{vap}}{\rho g} + \frac{v_{in}^2}{2g} \tag{31}$$

the absolute margin of the inlet head above vapour pressure. The standard performance criterion is NPSH_3, the available NPSH at which the developed head has dropped 3% below its fully wetted value at constant speed and flow [12]. Suction capability is compared across machines through the suction specific speed, defined like eq. (1) but with NPSH_3 in place of head.

2.2 Turbine Theory

2.2.1 Impulse Turbines

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Figure 5: Impulse turbine velocity triangle nomenclature at the rotor inlet (station 3) and exit (station 4). The design point triangles with actual values are shown in fig. 16.

Turbines work with the same Euler turbine equation. In the supersonic impulse turbine design, the blades are only occupy a small outer portion of the blisk, so the turbine equation can be simplified:

$$\Delta h_{\text{use}} = c_{3u}u_3 - c_{4u}u_4 = u(c_{3u} - c_{4u}) \tag{32}$$

From eq. (32), it is understood that a higher flow velocity difference $c_{3u} - c_{4u}$ per unit mass will lead to a higher Δh_0 according to equation eq. (6). This means that high absolute velocities, and for the maximum amount of the flow angled in the circumferential direction is optimal. For this reason, gas generator cycle turbines tend to take advantage of the high pressure ratios and thus the supersonic velocities. Then for the angles, the highest turning angle achievable is optimal. The amount of flow turning is usually limited by the achievable geometry.
The ideal efficiency of the impulse stage follows from eq. (32) and the velocity triangles. At the rotor inlet and exit the circumferential components of the absolute and relative velocities are related by

$$c_{3u} = w_{3u} + u, \qquad c_{4u} = u + w_{4u} \tag{33}$$

The ideal impulse blade turns the relative flow without slowing it down, so the exit relative velocity mirrors the inlet one, $w_{4u} = -w_{3u}$. Substituting into eq. (32),

$$\Delta h_{\text{use}} = u\,(c_{3u} - c_{4u}) = u\,[(w_{3u} + u) - (u + w_{4u})] = 2u\,w_{3u} = 2u\,(c_{3u} - u) \tag{34}$$

Two things come out of eq. (34). First, solving it for the inlet swirl gives the velocity triangle inversion used by the sizing procedure of section 3: given a required specific work and a blade speed, the nozzle must deliver

$$c_{3u} = \frac{\Delta h_{\text{use}}}{2u} + u, \qquad c_3 = \frac{c_{3u}}{\cos\beta_3}, \qquad c_{3m} = c_3 \sin\beta_3 \tag{35}$$

where β_3 is the nozzle angle measured from the circumferential direction. Second, dividing the work by the kinetic energy the nozzle supplies gives the stage efficiency. In a pure impulse stage the entire expansion happens in the nozzle, so the available energy per unit mass is the jet kinetic energy $c_3^2/2$, and with $c_{3u} = c_3 \cos\beta_3$,

$$\eta_{\text{ideal}} = \frac{2u\,(c_3\cos\beta_3 - u)}{c_3^2/2} = 4\,\frac{u}{c_3}\left(\cos\beta_3 - \frac{u}{c_3}\right) \tag{36}$$

The efficiency depends only on the blade to jet speed ratio u/c_3 and the nozzle angle. By taking the derivative of eq. (36) and setting it to zero, the optimum sits at $u/c_3 = \cos\beta_3/2$, where the exit swirl is zero and the efficiency peaks at $\cos^2\beta_3$. The same expression with the Weiss loss coefficients of section 2.2.3 attached becomes $\eta_h = 2k_N^2\,(u/c_3)\,(\cos\beta_3 - u/c_3)\,(1 + k_R)$, which reduces to eq. (36) when the nozzle and rotor are loss free ($k_N = k_R = 1$); this lossy form is the solid curve in fig. 6.

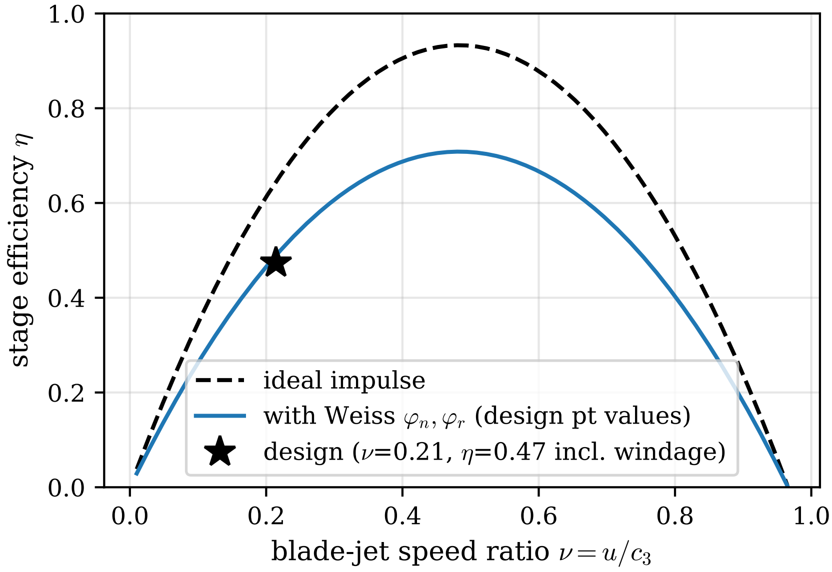


Figure 6: Impulse stage efficiency against blade to jet speed ratio. The dashed line is the ideal impulse stage of eq. (36), the solid line includes the Weiss loss coefficients evaluated at the design point, and the star marks the design point of this turbine, far below the optimum speed ratio.

The jet velocity C_0 used in this ratio is the spouting velocity, the velocity reached by an ideal isentropic expansion of the gas over the full stage pressure ratio,

$$c_s = \sqrt{2\,\Delta h_s} = \sqrt{2\,c_p T_{01}\left[1 - \left(\frac{p_e}{p_{01}}\right)^{\frac{\gamma-1}{\gamma}}\right]} \tag{37}$$

It links the available inlet pressure directly to the achievable jet velocity, and it is what makes the blade speed problem concrete. A small turbine on a 20,000 RPM shaft has a blade speed of around 100 m/s, while the spouting velocity of even a moderate pressure ratio gas generator is several hundred m/s, so U/C_0 is forced far below the optimum of around 0.5.
NASA special publication 8110 shows that for low U/C_0 turbines, impulse turbines are more efficient, even if reaction turbines can have a higher overall efficiency, just not at the low U/C_0 regime. In addition, the same conclusion is drawn by comparing the influence of tip clearance on the two types, where impulse turbines are more robust to tip clearance. This is especially important for small scale turbines where manufacturing tolerances are more difficult to achieve and contribute to more significant losses [**?**].

2.2.2 Blade Height and Partial Admission

Another problem with designing small turbines is with the low blade height required. By calculating the mass flux through the turbine, the blade height can be designed:

$$H_3 = \frac{\dot{m}}{\rho_3 c_{3m} d_{\text{mean}}} \tag{38}$$

Where ρ_3 is the density of the turbine gas at the turbine inlet. Using the example turbine above, the required blade height would be . By introducing partial admission, the mass flow through the turbine can be restricted to a smaller arc around the annulus at a slight cost to efficiency. The partial admission ratio ζ can be included into the blade height formula as so:

$$H_3 = \frac{\dot{m}}{\zeta \rho_3 c_{3m} d_{\text{mean}}} \tag{39}$$

By reducing the partial admission to , the blade height is increased to . The design of the turbine blade height is very important due to multiple reasons. Firstly, the aspect ratio of the turbine blade determines the magnitude of of the wall boundary layer blockage. For this reason, a aspect ratio between is desired.
Another way to increase the blade height is to decrease the meridonal velocity c_{3m}. This is achieved by increasing the angle that the flow has to turn as $c_{3m} = c_3 \sin(\beta_3)$.
The cost of partial admission and of low aspect ratio blading is not guesswork, as both have experimentally validated loss theories. Ohlsson derived and validated a closed form partial admission loss model where the efficiency penalty is a function of the ratio of active arc length to blade chord [14], and the same rig produced loss correlations for very low aspect ratio impulse blades [15]. These are used to check the loss model predictions in section 5.
For supersonic flow to develop within a turbine blade passage, a condition called "starting" must be satisfied. On the startup of a supersonic turbine, a normal shock starts at the inlet of the blade passage and as to pass through the whole passage before the flow into and out of the turbine passage can be supersonic.

2.2.3 Loss Model

The main loss model used in the design of the turbine is the one provided by Weiß [?]. There are more general loss models provided by texts such as NASA SP-290, giving loss models for tip-clearance, disk friction, partial admission and incidence losses. However, those models were built on large full-admission, subsonic to transonic turbines. Weiss specifically derived his loss model for the specialized application of a supersonic impulse turbines at small scales (1-200kW). As the regime of the turbine considered in this thesis matches closer to Weiss's model compared to SP-290's model, Weiss's model is used. Weiss describes different models lead to stage efficiency differences of about 10% points. Derived from rocket turbopumps literature and adjusted by in-house measurements, Weiss provides three efficiencies.

$$k_N = \sqrt{1 - (0.0029\, M_3^3 - 0.0502\, M_3^2 + 0.2241\, M_3 - 0.0877)} \tag{40}$$

eq. (40) is the nozzle loss coefficient, relating the actual velocity to the theoretical isentropic velocity by $k_N = c_{No}/c_{No,is}$.

$$\begin{aligned} k_R = 0.957 &- 3.62 \times 10^{-4}\, \Delta\beta - 2.58 \times 10^{-2}\, M_{w3} \\ &+ 6.39 \times 10^{-6}\, \Delta\beta^2 + 6.74 \times 10^{-2}\, M_{w3}^2 \\ &- 7.53 \times 10^{-8}\, \Delta\beta^3 - 4.30 \times 10^{-2}\, M_{w3}^3 \\ &- 2.38 \times 10^{-4}\, \Delta\beta\, M_{w3} + 1.45 \times 10^{-6}\, \Delta\beta^2 M_{w3} + 4.25 \times 10^{-5}\, \Delta\beta\, M_{w3}^2 \end{aligned} \tag{41}$$

eq. (41) is the loss of velocity along the blade passage through a theoretically impulse blade, relating the actual velocity to the theoretical velocity by $k_R = w_{exit}/w_{inlet}$.

$$P_v = \frac{1.85}{2}\,(1-\varepsilon)\,\rho_3 \left(\frac{n}{60}\right)^3 d_m^4\,(4.5\, b) \tag{42}$$

Finally, eq. (42) captures the losses due to partial admission. Weiss calls this loss ventillation losses, but it can also be called windage losses. Ventilation losses refers to the the non-admitted arc being dragged through stagnant air.
The choice of Weiss's model over the classical decompositions is a regime match rather than a claim that one is better. The losses that dominate a small supersonic partial admission impulse stage are supersonic friction in the nozzle and blade passage plus ventilation, which is exactly what Weiss's coefficients parameterise. The classical loss breakdowns such as Glassman's are anchored in large full admission machines where tip clearance and incidence dominate [16], and those losses are largely absent here by architecture, since the blades are shrouded and the velocity triangles are matched at the design point. Whether the Weiss prediction is actually trustworthy is checked against measured data from a machine of the same architecture in section 5.

2.2.4 Method of Characteristics Blade Design

The design so far is based on 1D mean line theory. The design of the actual blade profile which will turn the supersonic flow in the desired direction is often accomplished using an application of the Method of Characteristics to transition a uniform flow field into vortex flow. .

Blade terminology and vortex flow Geometry is separated into the turbine blade, and the passage between the flow. Two surfaces are separately designed, and it is common to refer to the top surface as the convex, and the bottom surface as the concave one. This is in relation to the solid blades themselves as opposed to an alternate designation by the top and bottom of the blade passage.
Vortex flow is a special type of supersonic flow in which flow follows a theoretical perfect arc around a central point. This is a very special type of flow field that is achieved by conditions where expansion and compression waves cancel each other to allow shockless rotation of the flow field. The vortex flow is bounded by the two circular arcs of the surfaces of which a desired surface Mach number is achieved. MOC generates the profile for desired mach numbers. Later analysis will need to be required to choose these desired Mach numbers.
A special condition of vortex flow satisfies the equation

$$\frac{V}{r} = \text{constant} \tag{43}$$

Boxer and Goldman nondimensionalizes this equation using the critical velocity $M^* = V/a^*$ where $a^* = \sqrt{\gamma R T^*}$ is the speed of sound at the sonic point. As a^* only depends on stagnation temperature, it is constant along any streamline of fixed stagnation enthalpy. Then, a nondimensional radius r^* is introduced as the sonic velocity streamline in the vortex field, giving the following condition:

$$\frac{M^*}{r^*} = 1 \tag{44}$$

The Prantdl-Meyer angle can be defined from the critical mach number M^* as:

$$\nu(M) = \frac{\pi}{4}\left(\sqrt{\frac{\gamma+1}{\gamma-1}} - 1\right) + \frac{1}{2}\left(\sqrt{\frac{\gamma+1}{\gamma-1}}\arcsin\left[(\gamma-1)M^{*2} - \gamma\right] + \arcsin\left(\frac{\gamma+1}{M^{*2}} - \gamma\right)\right) \tag{45}$$

And the converting between Mach number and Critical Mach number is given by:

$$M^* = \sqrt{\frac{(\gamma+1)M^2/2}{1+(\gamma-1)M^2/2}} \tag{46}$$

Equations eq. (44) and eq. (46) can be used to derive the nondimensional radius of the upper and lower surfaces of the blade. The end points of this circular arc is exactly the angle by which the inlet flow has to be turned to achieve the surface mach number.

$$\begin{aligned} \alpha_{l-i} &= \beta_1 - (\nu_i - \nu_l) \\ \alpha_{u-i} &= \beta_1 + (\nu_i - \nu_u) \end{aligned} \tag{47}$$

The equations for the outlet are the same as in eq. (47).

Transition arcs To transition from the uniform inlet flow to the vortex flow, method of characteristics is used to generate the blade profile to achieve this in a shockless manner.
To achieve this, the flow angles from β_i to α_i is discretized into k interval of ϕ_0 to ϕ_k. It is shown by that the flow angle and dimensionless radius R^* is related along its characteristic line with:

$$\phi = \pm\frac{1}{2}f(R^*) + \text{constant} \tag{48}$$

The positive gives the expansion line and the negative gives the compression line. Where $f(R^*)$ is:

$$f(R^*) = \sqrt{\frac{\gamma+1}{\gamma-1}}\arcsin\left(\frac{\gamma-1}{R^{*2}} - \gamma\right) + \arcsin\left((\gamma+1)R^{*2} - \gamma\right) \tag{49}$$

$f(R^*)$ for a given flow angle can be calculated with:

$$f(R^*_k) = 2\nu_i - \frac{\pi}{2}\left(\sqrt{\frac{\gamma+1}{\gamma-1}} - 1\right) - 2k\Delta\nu \tag{50}$$

The result of eq. (50) can be used to numerically solve for R^*_k with eq. (49), and points corresponding to each flow angle along the major expansion characteristic line can be calculated with:

$$\begin{aligned} x^*_k &= -R^*_k \sin(\phi_k) \\ y^*_k &= R^*_k \cos(\phi_k) \end{aligned} \tag{51}$$

The wall segments are generated by starting from the end point of the circular arc, then finding the intersection of the previous wall segment and the mach line from this point. The slope of the mach line with respect to to the x^* axis is given by:

$$m_k = \tan\left(\frac{\phi_k+\phi_{k-1}}{2} + \frac{\mu_k+\mu_{k-1}}{2}\right) \tag{52}$$

Where

$$\mu_k = -\arcsin\left(\frac{1}{M_k}\right) = -\arcsin\left(\sqrt{\left(\frac{\gamma+1}{2}\right)R_k^{*2} - \frac{\gamma-1}{2}}\right) \tag{53}$$

By iterating through the flow angles for both the upper and lower surfaces for the inlet and outlet, the full theoretical MOC blade profile can be generated.

2.2.5 Surface Mach Number Selection

There are two main factors to consider when choosing the inlet, outlet and surface mach numbers. One is to prevent flow separation, and the other is to ensure that supersonic flow within the blade passage can "start".

Flow separation Separation occurs in high adverse pressure gradients. In supersonic flow, this happens whenever the flow is decelerating along the surface. There are two main regions on the blade surfaces which separation can occur. At the upper surface, the inlet transition arc accelerates the flow from the inlet velocity until the vortex flow, then decelerates the flow until the outlet. At the lower surface, the inlet transition arc decelerates the flow from the inlet velocity until the vortex flow, then accelerates the flow until the outlet.
Goldman believes that separation on the lower surface is probably not as important as the flow would tend to reattach shortly as the pressure gradient becomes favourable again [?]. Goldman uses the value of the incompressible form factor H_i along the blade surface to predict separation, where Schlichting found that separation usually occurs in the range of H_i from 1.8 to 2.4.
To find H_i along both surfaces, Sasman and Cresci [?] provides a method to calculate the compressible turbulent integral boundary layer with two ODEs.

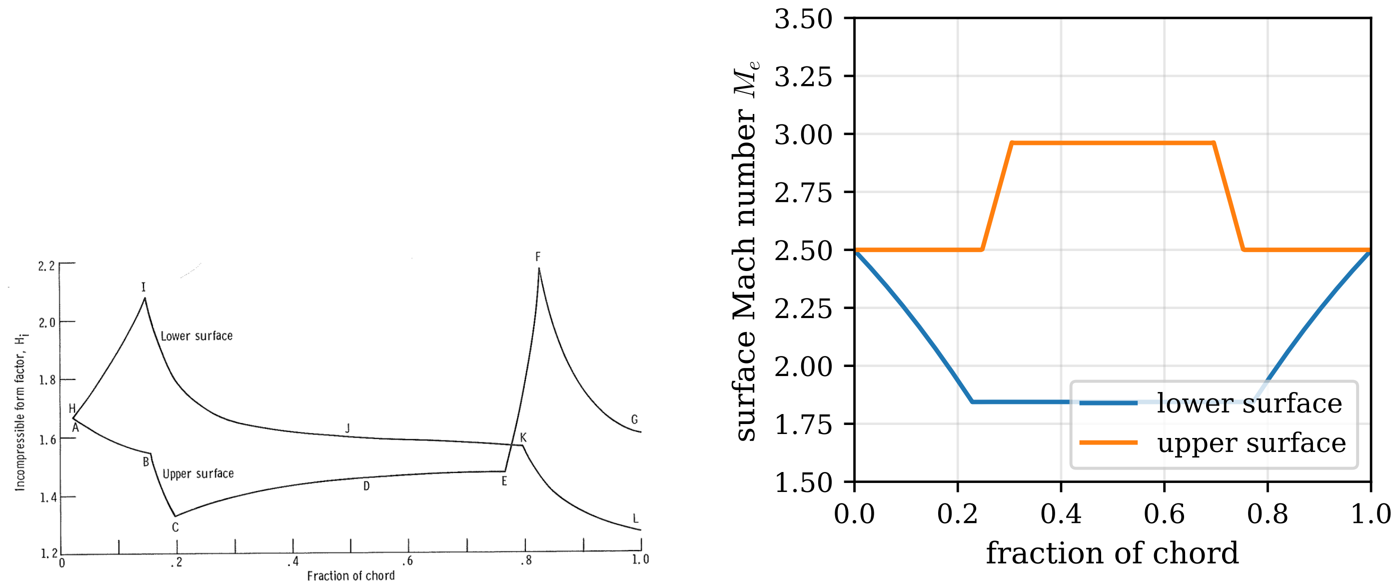
Once a MOC blade profile is generated and the mach number at every point along the blade is known, the momentum-integral equation and shape-factor equation can be solved as an initial value problem starting from the leading edge goverened by the following equations:

$$\frac{df}{d(s/c)} = 1.268\left[-\frac{f}{M_e}\frac{dM_e}{d(s/c)}\left(1+g_w H_i\right) + A\,c\right] \tag{54}$$

$$A\,c = 0.123\,e^{-1.561 H_i}\,M_e\,\mathrm{Re}_0\,\frac{T_e}{T_0}\left(\frac{\bar{\mu}}{\mu_0}\right)^{0.268} \tag{55}$$

$$\frac{dH_i}{d(s/c)} = \underbrace{-\frac{1}{2M_e}\frac{dM_e}{d(s/c)}\,H_i(H_i+1)^2(H_i-1)\left[1+(g_w-1)\frac{H_i^2+4H_i-1}{(H_i+1)(H_i+3)}\right]}_{\text{pressure-gradient term: deceleration drives } H_i \text{ up}} + \underbrace{\frac{H_i^2-1}{f}\,A\,c\left[H_i - \frac{0.011\,(H_i+1)(H_i-1)^2}{H_i^2\,(C_f/2)}\frac{T_e}{\bar{T}}\right]}_{\text{shear term: wall friction relaxes } H_i \text{ back toward equilibrium}} \tag{56}$$

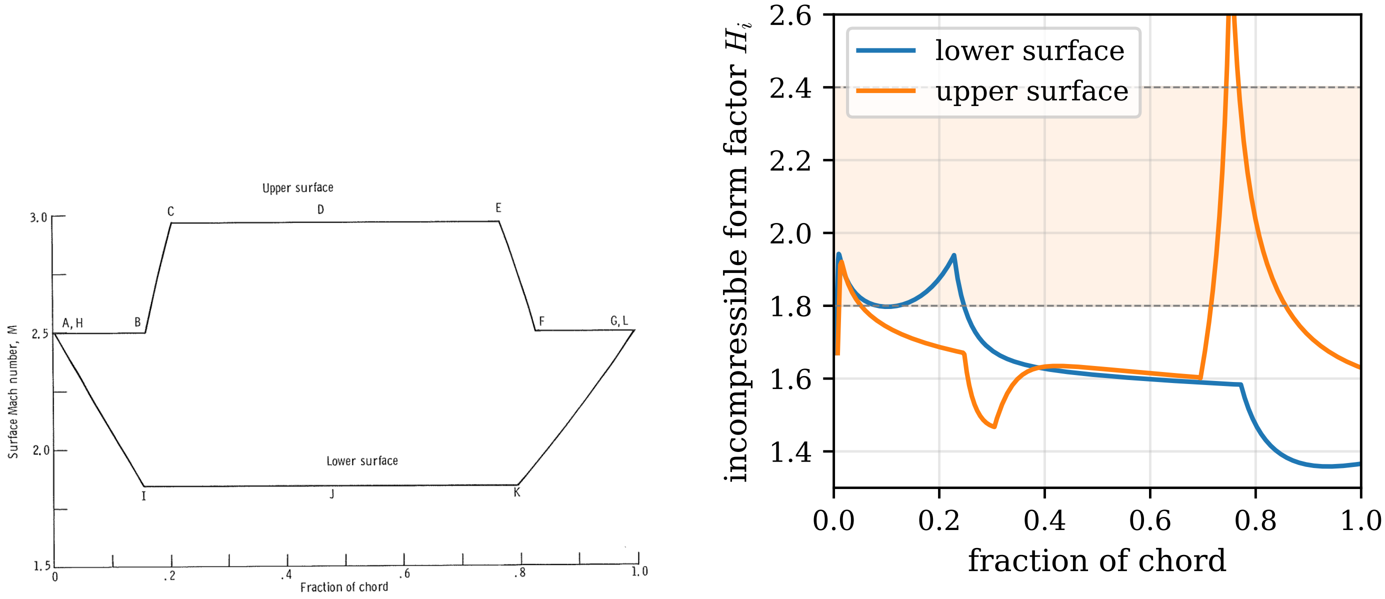
With initial conditions of a flat-plate $\frac{\theta_0}{c} = 0.036\,\frac{s_0}{c}\,\mathrm{Re}_{s_0}^{-1/5}$.
The initial condition for the shape factor is chosen so that it matches closely to the rapidly converging value $H_{i,0} = 1.67$.
It is important to note that Goldman provides equations in his NASA TN D-4421 1968 paper that has been shown to not be conservative enough, with tests by Erikson saying that his criteria is a little too optimistic. The method described here is from Goldman's NASA TM X-2095 paper.
Two limitations of the method should be stated, both flagged by Goldman himself [17]. The H_i threshold band of 1.8 to 2.4 is rooted in incompressible data, and the method assumes a turbulent boundary layer at a chord Reynolds number of 3.5×10^4 in his reference case, where laminar flow is plausible. The flight programme experience supports running the check on both surfaces, as the HM60 supersonic stage found boundary layer growth considerable with separation starting at around 25% chord on the convex side, which forced a rotor redesign [8].



(a) Goldman's published distribution [17]

(b) This work's implementation

Figure 7: Validation of the MoC implementation against Goldman's typical design case ($M_{in} = 2.5$, $\beta = 70°$). The surface Mach plateaus reproduce to within 1% (2.96 and 1.84 against Goldman's 2.97 and 1.85).



(a) Goldman's published form factor [17]

(b) This work's implementation

Figure 8: Validation of the Sasman and Cresci boundary layer solver against the same Goldman case. The input Mach distribution was reconstructed from the published design parameters. The peak H_i location and the separation verdict reproduce, while the peak magnitude overshoots (2.6 against Goldman's 2.2 at the same chord position).

Starting On startup before the turbine passage runs supersonically, a detached bow shock stands ahead of the leading edges. The flow behind this normal shock is subsonic and has lost a significant amount of total pressure. For the turbine to operate nominally with supersonic flow, the passsage must be able to "swallow" this shock, allowing the shock to move through the whole passage.

To do so, the do so, the post shock flow with a reduced total pressure must have enough of a flow area pass the required mass flow similar to the choked flow condition. A similar condition is the unstart of a supersonic wind tunnel or a supersonic internal intake. However, Boxer noticed in 1952 that due to the large velocity gradient between the channel walls due to tue turning, the flow cannot be assumed one directional as in the previous examples. Goldman in 1968 provides the condition accounting for the "reduction of mass flow due to curvilinear flow" to calculate the maximum inlet design mach number to ensure starting as:

$$\frac{p_{02}}{p_{01}}(M_i) \;\geq\; \frac{Q}{1-C} \tag{57}$$

Each term carries real content, so they are written out in full here because the off-design version of this inequality is what explains the coupled test in section 5. The left side is the total pressure recovery across the detached normal shock at the inlet design Mach number M_i, given by the Rankine and Hugoniot relation

$$\frac{p_{02}}{p_{01}}(M_i) = \left[\frac{(\gamma+1)M_i^2}{(\gamma-1)M_i^2+2}\right]^{\frac{\gamma}{\gamma-1}} \left[\frac{\gamma+1}{2\gamma M_i^2-(\gamma-1)}\right]^{\frac{1}{\gamma-1}} \tag{58}$$

which falls steeply as M_i rises, so a higher inlet Mach number leaves less recovered pressure to push the swallowed shock through. The right side is the post shock area demand. Q is the vortex flow parameter, the weight flow the curved passage would pass relative to a straight channel, written over the passage bounded by the lower and upper surface critical Mach numbers M_l^* and M_u^* (related to the true Mach numbers by eq. (46)) as

$$Q = \frac{M_l^*\,M_u^*}{M_u^*-M_l^*}\int_{M_l^*}^{M_u^*}\left(\frac{\gamma+1}{2}-\frac{\gamma-1}{2}M^{*2}\right)^{\frac{1}{\gamma-1}}\frac{dM^*}{M^*} \tag{59}$$

and C is the reduction in passable mass flow caused by the two dimensional velocity gradient across the passage that Boxer identified,

$$C = 1-\sqrt{\frac{\gamma+1}{\gamma-1}}\left(\frac{\gamma+1}{2}\right)^{\frac{1}{\gamma-1}}\frac{K_{max}^*\,M_u^*}{M_u^*-M_l^*}\,I_R, \qquad I_R = (1-K_{max}^{*2})^{\frac{1}{\gamma-1}}-\left[1-K_{max}^{*2}\left(\frac{M_u^*}{M_l^*}\right)^2\right]^{\frac{1}{\gamma-1}} \tag{60}$$

where K_{max}^* is the value that maximises the swallowed weight flow, found by solving the weight flow identity

$$\int_{M_l^*}^{M_u^*}\left[1-\left(\frac{K_{max}^*}{M_l^*}\right)^2 M^{*2}\right]^{\frac{1}{\gamma-1}}\frac{dM^*}{M^*} = (1-K_{max}^{*2})^{\frac{1}{\gamma-1}}-\left[1-K_{max}^{*2}\left(\frac{M_u^*}{M_l^*}\right)^2\right]^{\frac{1}{\gamma-1}} \tag{61}$$

Solving eq. (57) as an equality for M_i gives the maximum inlet design Mach number that will still start, which is the limit applied in the blade design of section 3. The two integrals eqs. (59) and (61) have no closed form and are evaluated numerically. The physical reading is the one used later: the limit is set by the surface Mach pair (M_l^*, M_u^*), so the same passage that is opened up for efficiency by spreading the surface Mach numbers is simultaneously pushed towards the starting limit, and at low blade speed the rising relative inlet Mach number crosses it.

Historically, starting was a major problem for supersonic turbines. Before the theory was established, Colclough had to build a special jig in which every blade could be pivoted to vary the contraction until the blade started.

Colclough's cascade tests also produced something the inviscid theory cannot, an empirical derating of the starting limit that accounts for boundary layer blockage. His measured startable contraction is $k = A_{min}/A_2 = 1.029 - 0.163(M_2 - 1)$, valid for $1.0 < M_2 < 1.4$ [18]. This is applied on top of the Goldman limit in the design of section 3, and the design inlet relative Mach number of this turbine sits inside its validity range.

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The efficiency trade The separation criterion limits the acceleration of the flow along the upper surface, and the deceleration of the flow along the lower surface. Then, the starting criterion provides an additional limit on the inlet design mach number based on these surface mach numbers.

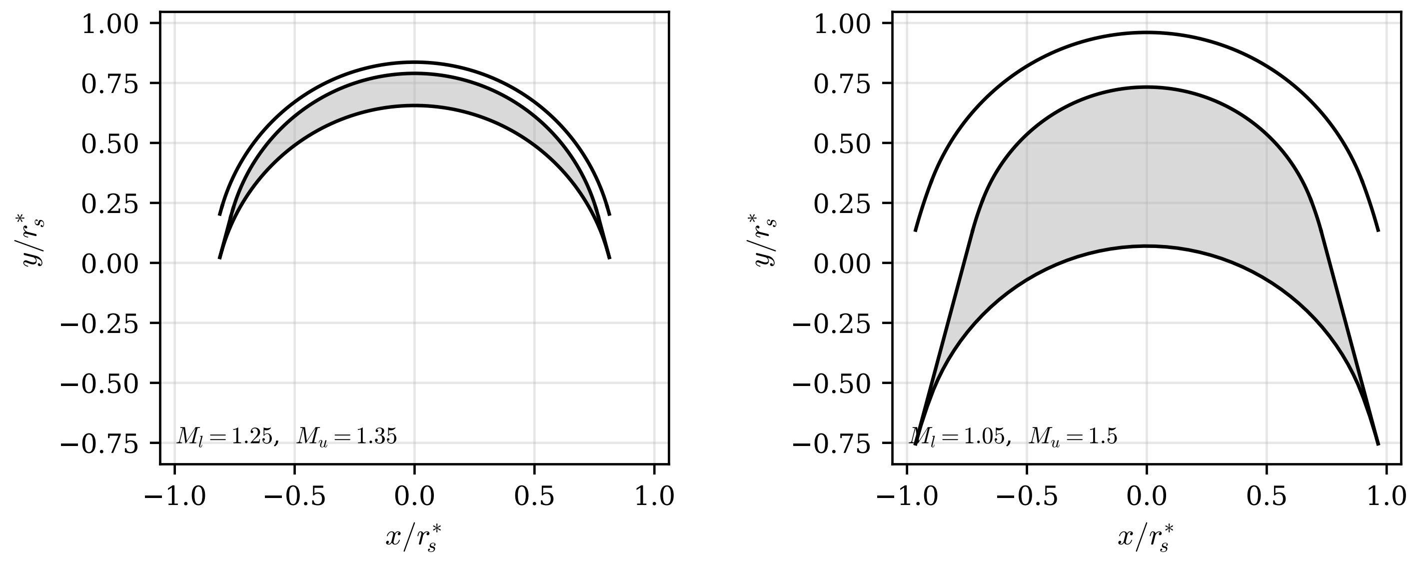
However, Goldman notes that for optimum efficiency, the maximum spread of surface number leads to maximum efficiencies. It comes from the obvious fact that a loss of velocity across the impulse blade leads to a loss of kinetic energy, so he posits the efficiency of an impulse blade to be:

$$\eta = \left(\frac{w_{exit}}{w_{exit,ideal}}\right)^2 \tag{62}$$

This is nearly the same as Weiss' eq. (41) loss. Weiss derived his empirical model only of 1D parameters such as inlet mach number and the angled the flow is turned.

What this efficiency tries to capture is the boundary layer friction losses, shocks, and exit wake mixing losses. The effect of spreading the surface mach numbers becomes clear by looking at the vortex equation eq. (44). The larger the mach number spread, the larger the difference in R^* between the upper and lower surface, the larger physical passage width. The wider the channel compared to the blade chord, the lower fraction of the boundary layer thickness to passage width, the lower the wetted surface area per unit flow, and thus the lower the losses.

This is the challenge of supersonic turbine blade design, maximum efficiency is achieved by riding along the edge of separation and starting.



(a) $M_l = 1.25$, $M_u = 1.35$

(b) $M_l = 1.05$, $M_u = 1.5$

Figure 9: The effect of surface Mach number spread on the blade passage, for the same inlet conditions and shared axis scales. A narrow spread produces a sliver of a passage where the boundary layers occupy a large fraction of the width, while a wide spread opens the passage up. This is why efficiency rises with Mach spread until separation or starting limits are hit.

2.2.6 Finite Leading Edge

In section 2.2.4, the upper and lower surface are mathematically designed to produce no shocks in transitioning the unifrom flow from the nozzle exit into the vortex flow. For no shocks to form, both surfaces must be perfectly parallel to the starting flow. This is problematic because this implies the point where the two surfaces intersect at the leading edge, the angle between the upper and lower surfaces is zero. This creates an impossible to manufacture zero thickness leading edge. Boxer suggests a few methods to solve this issue. He separates design into turbines with a subsonic or supersonic axial velocity component.

For subsonic axial velocity turbines, either lower or upper surface can be designed with an incidence angle to the inlet flow. Angling the upper convex surface and keeping the lower concave surface parallel to the inlet flow, the compressive oblique shock will be generated along the region between the inlet and the nozzle exit, affecting the flow in the next blade passage. Kantrowitz shows that in steady state a compression shock of equal strength will be created to obtain the steady state condition [**?**]. To solve this, a small angle wedge can be placed shortly after the leading edge to generate a small expansion wave to cancel out the shock.
Another method is angling the lower concave surface so that the shock stays within the blade passasge and does not affect the upstream flow. The oblique comrpession shock can then be designed to intersect the upper surface at an angle to fully cancel out the shock.
For supersonic axial velocity turbines, a wedge can be designed on the convex surface with a low enough angle from the design inlet flow so that the shock stays within the blade passage and can be cancelled out by a kink on the concave surface.

figu exp this

Figure 10: Finite leading edge treatments for supersonic blades: incidence angling of one surface, in passage shock cancellation, and the small wedge used to cancel the resulting compression shock.

2.2.7 Nozzle Design

The nozzle throat sets the stage mass flow. Once the pressure ratio exceeds the critical value the throat chokes, and the mass flow depends only on the throat area and the upstream stagnation state:

$$\dot{m} = \frac{A_{th}\, p_{01}}{\sqrt{T_{01}}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{63}$$

This equation does double duty in this thesis. In design it sizes the nozzle throats for the required mass flow, and in the analysis of section 5 it is the only available estimate of the air mass flow, since the rig carries no air flowmeter. Its key property is that it is independent of shaft speed and of everything downstream of the throat, which makes it valid whether or not the rotor is operating as intended.
While the throat sets the mass flow, the ratio of the exit area to the throat area sets the exit Mach number, through the isentropic area relation

$$\frac{A_3}{A_{th}} = \frac{1}{M_3} \left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2} M_3^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{64}$$

There is no closed form for M_3, so eq. (64) is inverted numerically for the supersonic root given the design area ratio. This relation is what links the nozzle geometry to the velocity triangle: the exit Mach number it returns must match the M_3 demanded by the inlet swirl of eqs. (35) and (70), and the same relation is solved again when the design is re-evaluated for a different working gas, since the area ratio is fixed by the printed hardware but the Mach number and exit velocity are not.
Historically, the optimal nozzle design for a ideal inviscid gas was designed with the Method of Characteristics as well. Rao provides a method to achieve this [**?**]. NASA special pubplication 8120 explains that a single canted parabola closely approximates the ideal nozzle contour and should be used for high performance nozzles. In this paper, the nozzle was a simple conical nozzle with a 7 degree half angle. It can be seen from that the performance of a conical nozzle for low area ratios lead to a efficiency loss below 2%.

add ure fror sp8

Figure 11: Conical against contoured nozzle efficiency as a function of area ratio, reproduced from NASA SP-8120. At the low area ratio of this design the conical penalty is below 2%.

2.2.8 Turbopump Matching

When the pump and turbine are coupled on one shaft with no speed control, the system settles where the torques balance:

$$\tau_{\text{turbine}}(\omega) = \tau_{\text{pump}}(\omega) + \tau_{\text{friction}}(\omega) \tag{65}$$

The two sides have opposite slopes. The pump torque rises with speed, approximately with ω^2 from the affinity laws. The turbine torque falls with speed, because for a fixed supply the jet velocity is fixed, so increasing blade speed reduces the velocity difference that generates the blade force. The intersection of the two curves is the match point. Because the turbine curve falls and the load curve rises, a small disturbance in speed produces a net torque that pushes the shaft back towards the intersection, so the match point is self regulating and no active speed control is needed. This is the behaviour the coupled test in section 5 is designed to demonstrate.

3 Design

The design point for this thesis was chosen from the capabilities of the test rig. The design parameters for the pump and turbine is shown in table 1 and this section explains how each number was chosen.
The RPM limit was set by the maximum speed of the Magtrol TM306 inline torquemeter and the AHB-3 hysteris brake at 20,000 RPM. The pressure rise and flow rate was chosen to study the low specific regime. The supply pump was decided to be Hozelock's JET 3000 K7 Jet pump as it was available in the department and had a published performance curve. At 0.3 kg/s, the supply pump had a head of around 2.2 m. Then, the head rise of the pump was chosen to be 20 bar for a specific speed of 6.4.

Table 1: Turbopump system design point.

Symbol	Parameter	Value
n	design shaft speed	20 000 rpm
Q	pump design volume flow rate	0.3 L/s
H	pump design head rise	204 m
Δp	pump design pressure rise	20 bar
n_q	pump specific speed (rpm, m^3/s, m)	6.4
P	turbine design shaft power	2.6 kW
p_{01}	turbine design inlet total pressure	7.65 bar
–	pump working fluid	water
–	turbine working fluid	air

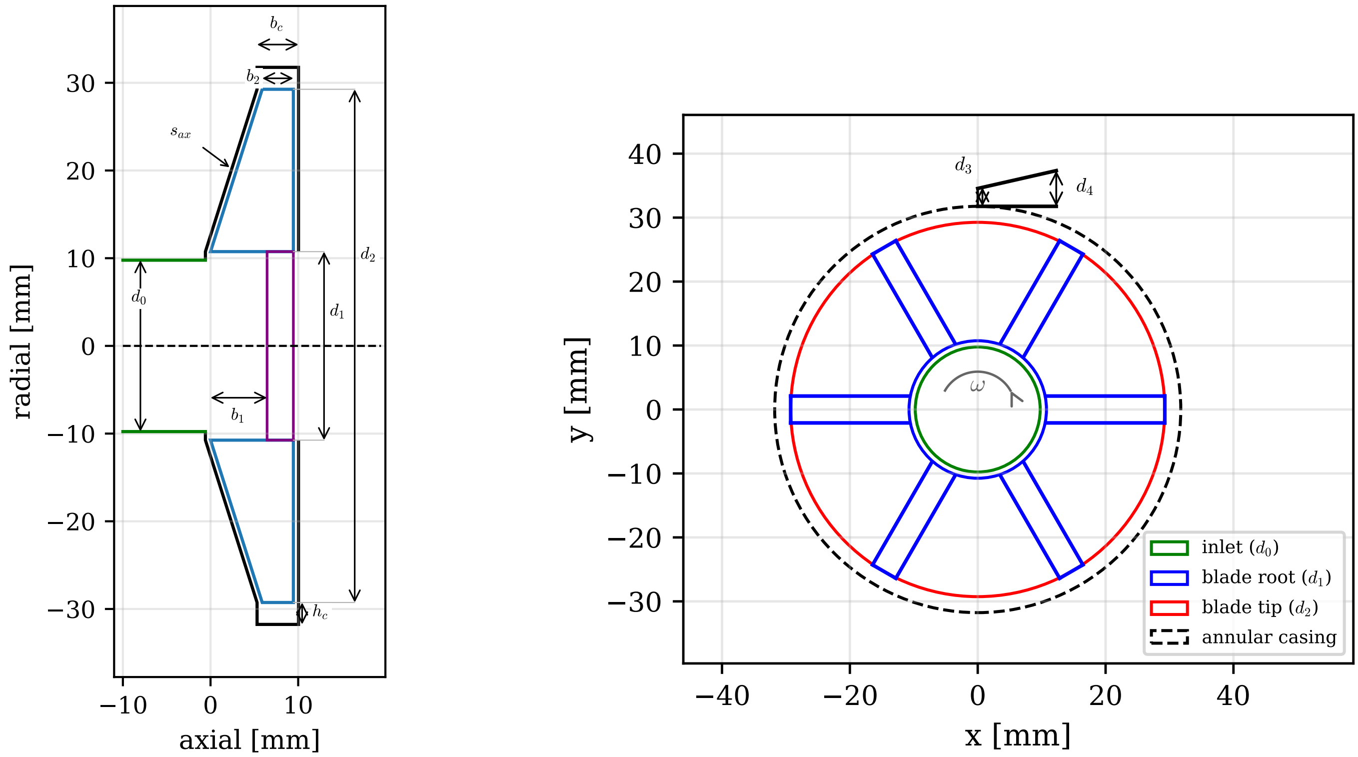
3.1 Pump Design

3.1.1 Design Point and Requirements

The pump design point is 0.3 L/s of water at a head rise of 204 m at 20 000 rpm. Substituting these into eq. (1) gives $n_q = 6.4$, which confirms the pump sits in the partial emission regime established in fig. 1. On top of the hydraulic requirements, the design had to respect the hardware limits of the rig. The AHB-3 hysteresis brake is rated to 1800 W, which caps the allowable shaft power, and the available NPSH is set by the supply pump and tank rather than being a free choice.

3.1.2 Sizing

The pump was sized with the Barske procedure of eqs. (11) and (12) and the accompanying guidelines, executed in order with this design point. A reasonable inlet velocity gives an eye diameter of $d_0 = 19.54\,\mathrm{mm}$, enlarged to $d_1 = 21.5\,\mathrm{mm}$ for the blade leading edge to allow the deliberate prerotation Barske recommends. The blade heights follow as $b_1 = 6.45\,\mathrm{mm}$ and $b_2 = 3.554\,\mathrm{mm}$. The tip diameter comes from the recovery equation eq. (12) with a deliberately conservative recovery coefficient of $c_k = 0.2$, giving $d_2 = 58.5\,\mathrm{mm}$. A conservative c_k oversizes the impeller slightly, and the plan was to trim d_2 later if the discharge pressure came out too high. The resulting geometry was checked against the constraints: the inlet blade speed stays below 45 m/s, $d_2/d_1 > 1.5$, the NPSH margin against the supply system is positive, and the predicted shaft power of 2567 W sits under the 1800 W brake limit.
It should be stated clearly that the sizing used the Barske model, not Lock's. Lock's refined model was adopted later as the analysis model because it predicts the drooping head curve and the cavitation cutoff that the Barske model cannot, and its comparison against the measured data is in section 5. Rearranging Lock's model into a direct sizing tool is left as future work.



(a) Meridional section

(b) End view

Figure 12: The sized pump geometry, drawn to scale from the sizing code output. The meridional section shows the open tapered blades, the conical front casing maintaining the axial clearance s_{ax}, and the annular casing of width b_c and depth h_c. The end view shows the six radial blades and the single tangential conical diffuser with throat d_3 and exit d_4. Values are listed in table 2.

Table 2: Pump design summary, generated directly from the sizing code.

Symbol	Parameter	Value
d_1	impeller inlet (blade leading-edge) diameter	21.5 mm
d_2	impeller tip diameter	58.5 mm
b_2	blade height at impeller exit	3.554 mm
z	blade count	6
d_3	diffuser throat diameter	3.0 mm
ψ	design head coefficient	1.065
n_q	specific speed (rpm, m^3/s, m)	6.4

3.1.3 Diffuser

The diffuser is a single conical passage placed tangentially on the annular casing, following Barske's guidance from section 2.1.4. The throat was sized for a throat velocity of 0.8 to 0.9 of u_2, giving $d_3 = 2.79\,\mathrm{mm}$, and the exit area is 3 to 4 times the throat area, giving $d_4 = 5.58\,\mathrm{mm}$. The throat is one of the two performance critical tolerances of the whole pump, because the entire flow passes through this one small hole and, as Lock's model predicts, the maximum flow rate is set by cavitation at exactly this location. The comparison of this throat sizing against the literature is discussed in section 5.

3.1.4 Mechanical Design

The dominant mechanical load is not torque but radial thrust. Because the pump is partial emission, the pressure field around the impeller is asymmetric, and the net radial force this produces governs the shaft and bearing design rather than efficiency considerations. The radial thrust is estimated with the standard projected area form

$$F_R = K_R \, \rho g H \, d_2 b_2 \tag{66}$$

where K_R is an empirical thrust coefficient and $d_2 b_2$ is the projected impeller discharge area [12]. For a single tangential collector the unbalance is large compared to a volute pump, and at the design head this force, not the 1.2 Nm of torque, is what loads the bearings. The shaft is a 10 mm solid stainless steel stub, chosen over printed or aluminium options to keep the rotor stiff enough to prevent rotor to stator rubs at the small axial clearance.

The mechanical seal sets its own requirement, and its friction was modelled rather than guessed because it sits between the torquemeter and the impeller, inside every torque measurement of section 5. The net closing pressure on the seal faces is the sealed pressure acting on the hydraulic unbalance of the face, plus the spring:

$$p_f = \Delta p \, (b - k) + p_{sp}, \qquad b = \frac{d_o^2 - d_b^2}{d_o^2 - d_i^2} \tag{67}$$

where b is the seal balance ratio built from the face outer and inner diameters d_o, d_i and the balance diameter d_b, $k \approx 0.5$ is the pressure profile factor for the fluid film between the faces, and p_{sp} is the spring force over the face area [12]. The friction torque and heat follow as

$$\tau_{seal} = p_f \, A_f \, r_m \, f, \qquad P_{seal} = \tau_{seal} \, \omega \tag{68}$$

with A_f the face contact area, r_m the mean face radius and f the face friction coefficient. The fitted single spring seal used here has a balance ratio of $b = 1.20$, an unbalanced seal, so the closing pressure grows faster than the sealed pressure. With the dry running textbook coefficient $f = 0.07$, the model predicts about 0.4 Nm and 860 W of face heating at the 20 bar, 20,000 RPM design point, and about 0.08 Nm and 60 W at the achieved test conditions. The design point heat flux would flash the water film at the faces to steam without an active flush, so the seal is flushed by the pumped water itself. Two parts of this model are tested against data later: the static starting torque, quoted by Karassik at 3 to 5 times the running torque, is compared against the measured stiction band in section 5, and the friction coefficient itself is fitted from the measured loss budget, which returns f well below the dry value because the faces run on a water film.

Rolling bearing friction was estimated with the SKF four component model, which sums a load dependent rolling term, a sliding term and, for a submerged bearing, a drag term:

$$M_{brg} = M_{rr} + M_{sl} + M_{drag} = \phi_{ish} \, \phi_{rs} \, G_{rr} \, (\nu_l \, n)^{0.6} + G_{sl} \, \mu_{sl} + M_{drag} \tag{69}$$

where the G factors carry the bearing geometry and load, the ϕ factors are shear heating and starvation corrections, and ν_l is the lubricant kinematic viscosity [19]. For the 61900 deep groove bearings running submerged in water, the model predicts roughly 0.2 W per bearing at test speed and under 4 W at the full design speed. Bearing friction is therefore negligible against the seal and churning losses, a conclusion the shutoff loss decomposition of section 5 supports.

The front casing and the diffuser were printed in clear resin and polished. This was a deliberate design decision, enabled by the low test pressures, so that cavitation could be observed directly through the casing wall. The payoff of this decision is the cavitation visualisation in section 5, which turned out to be one of the most valuable measurements of the project.

The pump shaft critical speed was bounded with the same overhung rotor estimate used for the turbine shaft, eq. (76). Treating the 10 mm stainless shaft ($E = 193$ GPa, $I = \pi d^4/64 = 4.9 \times 10^{-10}$ m^4) as a cantilever from the bearing with the impeller mass at the overhang, the first bending speed stays above approximately 45,000 RPM for any plausible combination of overhang length up to 50 mm and impeller mass up to 100 g, a margin of better than 2.2 over the design speed. The bound is deliberately conservative, since the short stub geometry was chosen exactly to keep this margin large, as discussed in section 4.

Figure 13: Static FEA of the impeller and shaft at 20 000 rpm showing peak stress and deflection.

3.1.5 Cavitation Prediction

At the design flow rate, the predicted NPSH required is 3.1 m. Compared against the NPSH available from the pressurised supply, this leaves a positive but not generous margin, which is one of the reasons the cavitation behaviour was made a primary measurement objective. The full NPSH against flow rate characterisation, and the visual confirmation of where cavitation actually occurs, are presented in section 5.

3.1.6 Manufacturing and As-Built

The impeller, casing and diffuser were printed by SLA. Additive manufacturing is acceptable for this pump precisely because of the forced vortex operation described in section 2. With the fluid rotating as a solid body, the relative velocity over the blade surfaces is low, so blade surface finish has little effect on performance and the print finish is sufficient. Two tolerances dominate performance and were flagged for inspection: the axial clearance s_{ax} between the blades and the casing, and the diffuser throat diameter. The as-built throat turned out to matter a great deal, as the true minimum area was found to be the 3.0 mm semicircular passage into the annular casing rather than the 3.8 mm cone cutwater downstream, and this distinction becomes central to the cavitation analysis in section 5. Full as-built deviation tables are in the appendix.

Figure 14: CAD section view of the as-built pump assembly, showing the impeller, transparent front casing, tangential diffuser and mechanical seal.

3.1.7 Comparison to Conventional Design

Before committing to the Barske architecture, a conventional shrouded impeller was carried through a full Kaplan method blade construction at this duty point. The attempt is documented in section A and its conclusion is what justifies the architecture choice. The design correlations repeatedly broke down outside their validity ranges, and the geometry that emerged had twisted multi streamline blades with sub millimetre closed channel widths, which is not manufacturable at this scale with any process available to this project. This is consistent with the published conventional design at the same specific speed, which required 0.31 mm closed channels [2]. The quantitative comparison of the two architectures against literature is made with the measured data in section 5.

3.2 Turbine Design

3.2.1 Design Point and Regime Shift

Turbine sizing has to satisfy many requirements simultaneously, many design variables. After the pump was sized for 20 000 rpm at a power of 2.6 kW, there are many remaining free variables.
The main parameters being turbine diameter D_mean, mass flow through the turbine $\dot{m}$, blade inlet angle β_3.
One thing has to be kept honest throughout this section and the results. The turbine has two operating regimes. The design point is 2.6 kW at 20 000 rpm on 7.65 bar supply. The test point is what the laboratory shop air could actually deliver, around 2.9 bar absolute at the stator with the pressure drooping under flow, which settled the coupled shaft at around 5400 RPM. table 3 compares the two. The choked supersonic nozzle behaves the same in both regimes because choked flow is speed independent, but the rotor stage similitude parameters were not matched at test, and this difference drives most of the discussion in section 5.

Table 3: Turbine design point against the achieved test point. The test supply pressure drooped under flow, so the quoted value is representative.

Parameter	Design	Test
Shaft speed	20 000 rpm	≈5400 RPM
Inlet total pressure	7.65 bar	≈2.9 bar
Inlet total temperature	300 K	≈308 K
Rotor inlet relative Mach M_{w3}	1.33	1.58
Blade to jet speed ratio	0.214	≈0.06
Rotor state	started	unstarted

3.2.2 Architecture Selection

The stage is an impulse stage, for the reasons established in section 2: at low blade to jet speed ratio the impulse stage is more efficient than a reaction stage, and it is far more tolerant of tip clearance, which matters when the rotor is printed [20]. The central constraint is diameter. Running the stage at the optimum blade to jet speed ratio on a 20,000 RPM shaft would need a mean diameter in the hundreds of millimetres, which does not fit the rig or the shaft dynamics, so the turbine accepts a small diameter and a heavily off optimum speed ratio instead. The small mass flow then cannot fill the annulus at any sensible blade height, so partial admission is used. The degree of admission $\varepsilon = 0.15$ was chosen by working eq. (39) backwards, picking the admission fraction that raises the blade height to 4.0 mm, the smallest height that prints reliably and keeps the aspect ratio sensible.

3.2.3 Sizing

The mean line sizing is the inverse design chain of the theory chapter, executed in order. The inputs are the shaft power P, speed n, mean diameter d_m, mass flow $\dot{m}$, nozzle angle β_3, degree of admission ε, the gas properties and the exit pressure p_e. The blade speed follows from the speed and diameter, $u = \omega d_m/2$, and the required specific work from the power, $\Delta h_{\text{use}} = P/\dot{m}$. The velocity triangle inversion eq. (35) then fixes the jet the nozzle must deliver, c_{3u}, c_3 and c_{3m}. The nozzle exit static state follows from conservation of energy in the adiabatic nozzle and the isentropic relations:

$$T_3 = T_{01} - \frac{c_3^2}{2c_p}, \qquad a_3 = \sqrt{\gamma R T_3}, \qquad M_3 = \frac{c_3}{a_3} \tag{70}$$

$$p_3 = \frac{p_{01}}{\left(1 + \frac{\gamma-1}{2} M_3^2\right)^{\frac{\gamma}{\gamma-1}}}, \qquad \rho_3 = \frac{p_3}{R T_3} \tag{71}$$

Because the stage is impulse, the rotor takes no static pressure drop, so the nozzle exit static pressure must equal the exit pressure p_e. The supply pressure p_{01} is therefore not free: it is iterated until $p_3 = p_e$, which is how the design tool arrives at the required stator supply pressure of 7.65 bar rather than assuming it. The blade height then follows from continuity through the admitted arc, eq. (39), and the total nozzle throat area from the choked flow relation eq. (63). Finally the Weiss loss chain of section 2.2.3 is wrapped around this ideal chain in an outer loop: the internal power is raised until the shaft power net of the rotor loss and ventilation meets the 2.6 kW target, so the quoted supply pressure already carries the losses. With the chain in place, the remaining free choices are the mean diameter and mass flow. These were chosen by sweeping the $(d_m, \dot{m})$ plane with the full model and overlaying the manufacturing and rig constraints: a minimum blade height of 4 mm for reliable printing, a minimum nozzle throat width of 0.5 mm, and a nozzle exit Mach ceiling of 1.8 to keep the rotor inside the validated range of the loss correlations. fig. 15 shows the swept design space, the feasible region, and the selected design point. The chosen $d_m = 95$ mm and $\dot{m} = 0.0428$ kg/s sit inside the feasible region with the blade height constraint binding, which is the quantitative version of the statement that the blade height, not the aerodynamics, sizes this machine.

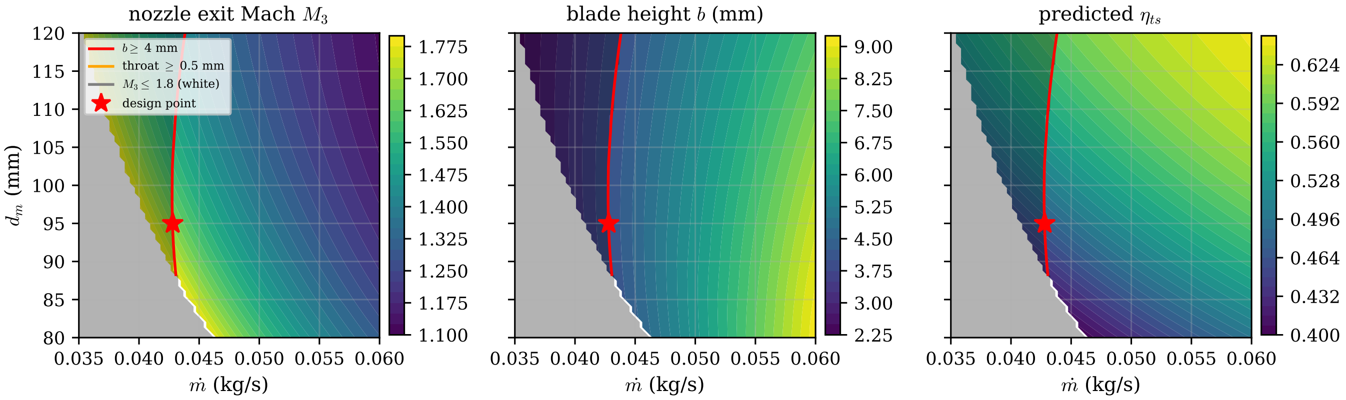


Figure 15: The turbine design space over mean diameter and mass flow, evaluated with the full sizing and loss model at the design speed. The panels show nozzle exit Mach number, blade height and predicted total to static efficiency. Overlaid lines are the design constraints (blade height $\geq$ 4 mm, nozzle throat $\geq$ 0.5 mm, $M_3 \leq 1.8$), the shaded region is infeasible, and the marker is the selected design point.

The mean line sizing gives a mean diameter of 95 mm, a blade speed of 99.5 m/s at design speed, a nozzle exit velocity of 464 m/s at Mach 1.67, and a rotor inlet relative Mach number of 1.33. The full output is in table 4 and the design point velocity triangles are drawn to scale in fig. 16. The triangles make the off optimum operation visible directly, as the blade speed is small next to the jet velocity and the exit swirl is strongly negative, which is the kinetic energy the stage fails to extract at this speed ratio.

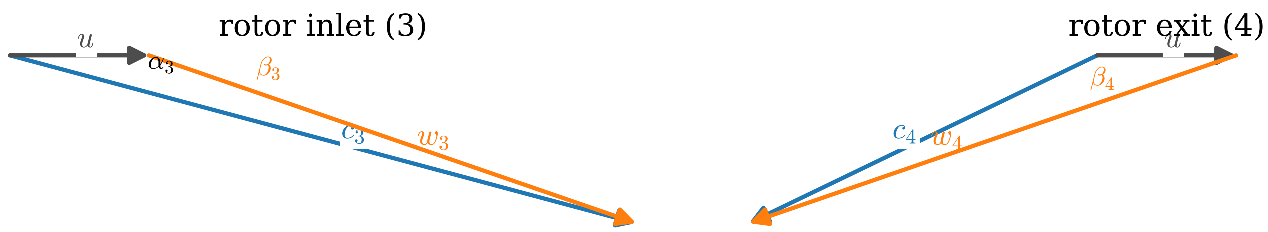


Figure 16: Design point velocity triangles at the rotor inlet and exit, drawn to scale from the sizing output. The exit triangle is the ideal symmetric impulse case. The large negative exit swirl is the direct consequence of operating at a blade to jet speed ratio of 0.214, far below the optimum.

Table 4: Turbine design summary, generated directly from the design code.

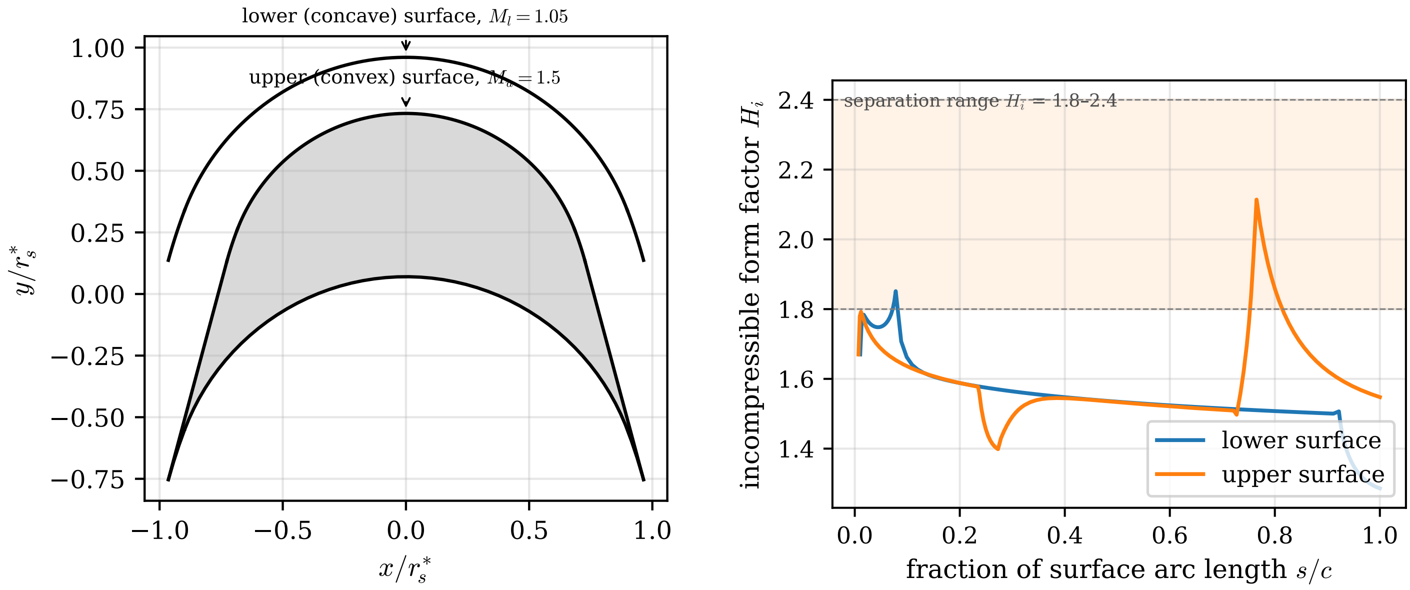
Symbol	Parameter	Value
P	design shaft power	2.6 kW
p_{01}	design inlet total pressure	7.65 bar
T_{01}	design inlet total temperature	300 K
$\dot{m}$	design air mass flow rate	0.0428 kg/s
d_m	mean (pitch) diameter	95 mm
b	rotor blade height	4.0 mm
ε	degree of admission	0.15
β_3	rotor blade angle (from tangential)	15°
z_N	number of nozzles	3
A_{th}	total nozzle throat area	35.5 mm^2
c_3	design nozzle-exit velocity	464 m/s
M_3	design nozzle-exit Mach number	1.67
M_{w3}	design rotor-inlet relative Mach number	1.33
η_{ts}	design total-to-static efficiency	0.47

3.2.4 Nozzle

The ideal nozzle contour for this application would be a Rao or canted parabola design as described in section 2. Due to time constraints and the negligible performance difference at this low area ratio, a simple conical nozzle with a 7 degree half angle and rounded corners was used instead. NASA SP-8120 puts the efficiency penalty of a conical contour below 2% at low area ratios, which is small against the other losses in this stage, so this is an honest engineering trade rather than a compromise of the results. There are 3 nozzles with a total throat area of 35.5 mm^2.

3.2.5 Blade Passage Design

The blade passage was generated with the MoC procedure of section 2.2.4 for an inlet relative Mach number of 1.3 at a blade angle of 75 degrees from axial, with surface Mach numbers $M_l = 1.05$ and $M_u = 1.5$. The choice of this surface Mach pair is the central design decision of the blade. The pair is not free: their mean is fixed near the inlet Mach number and their spread sets the turning the stage needs. The lower bound is separation on the pressure surface, checked with the Sasman and Cresci method of section 2, and the upper bound is starting, checked with the Goldman criterion derated by the Colclough fit. Following Goldman's observation that efficiency rises with Mach spread, the spread was opened until separation became imminent. fig. 17 shows the result: the as-designed blade with the H_i check touching 2.1 at around 77% chord on the upper surface, deliberately separation imminent at the design point. The boundary layer displacement correction applied to the printed contour is shown in fig. 18.
The starting margin was left deliberately thin to maximise loading. The Goldman limit for this passage is $M_{w3} = 1.408$ against a design value of 1.33. The same calculation run off design predicted, before any coupled test took place, that the rotor cannot start below approximately 15 300 RPM, because at low blade speed the relative inlet Mach number rises above the limit. This prediction is a design stage statement. What happened when the coupled test ran far below this threshold is the subject of section 5.



(a) As-designed blade passage

(b) Separation check

Figure 17: The as-designed MoC blade passage and its separation check. The form factor H_i is evaluated on both surfaces at the design point inlet conditions and touches 2.1 at around 77% chord on the upper surface, at the bottom of the 1.8 to 2.4 separation band. The blade is separation imminent by design, which per Goldman is where the efficiency optimum sits.

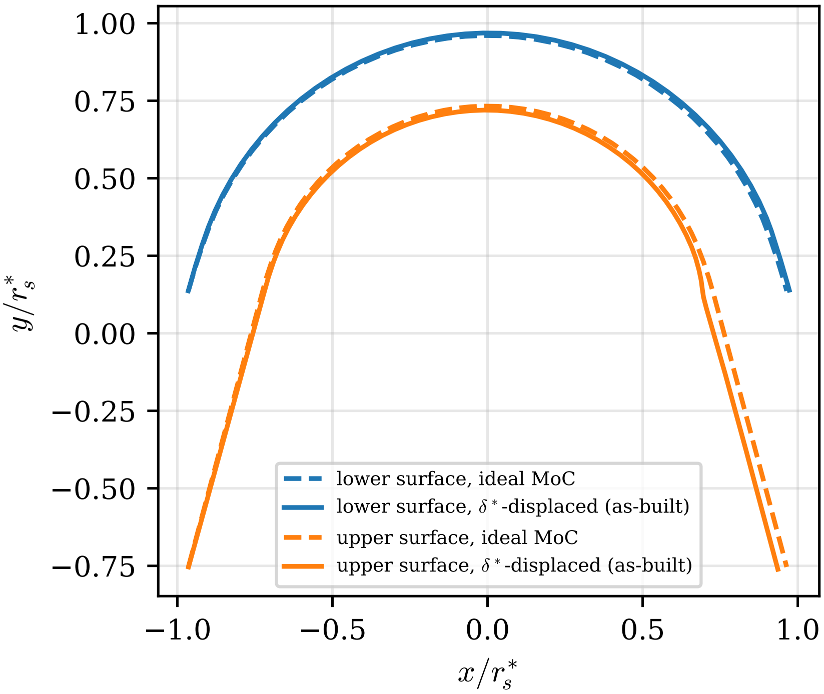


Figure 18: Ideal MoC contour (dashed) against the boundary layer displaced contour that was actually manufactured (solid). The surfaces are offset by the local displacement thickness δ^* from the Sasman and Cresci solution, so that the printed passage presents the inviscid design geometry to the core flow.

3.2.6 Losses and Predicted Efficiency

Applying the Weiss loss chain of section 2.2.3 at the design point gives a nozzle velocity coefficient of $k_N = 0.917$, a rotor velocity coefficient of $k_R = 0.807$, and a ventilation power of 85 W. Together these take the ideal stage efficiency of around 65% down to a predicted total to static efficiency of 0.47. As one external sanity check, the roughly 28 percentage point debit from ideal that this chain predicts matches the debit measured experimentally on a machine of the same architecture [21], which is examined properly in section 5.

3.2.7 Blade Geometry Details

Three geometry details depart from the ideal MoC contour. First, the leading edge. The MoC construction produces a zero thickness leading edge, which cannot be printed. A finite wedge of about 5 degrees was added on the pressure surface following the approach in section 2. The shock detachment angle at the design relative Mach number of about 1.3 is around 6.5 degrees, so a 5 degree wedge keeps the oblique shock attached. Second, the trailing edge is truncated to a flat of about 0.8 mm so it survives both the shock loading and embrittlement, since the expansion cools the air to around −88 °C at the exit. Third, balance holes are drilled through the disc web to equalise the pressure across the rotor and remove around 400 N of parasitic axial thrust. Balance holes and pistons are standard rocket pump practice [22], and a CFD study of a Barske type pump in the same regime showed balance holes reducing the axial force from 17 kN to 350 N, the same order of effect as needed here [1]. Finally, the blades are shrouded, which reduces the blade tip deflection by a factor of around 48 and removes tip leakage as a loss mechanism.

3.2.8 Cold Flow Thermal Environment

Expanding room temperature air to Mach 1.67 cools it to a static temperature of around 185 K, and the printed resin blades have to survive this. The static temperature is however not what the blade surface sees. A surface in high speed flow recovers part of the kinetic energy as frictional heating in the boundary layer, and equilibrates at the adiabatic wall temperature

$$T_{aw} = T_3 + r\,\frac{w^2}{2c_p}, \qquad r = \mathrm{Pr}^{1/3} \tag{72}$$

where r is the recovery factor for a turbulent boundary layer and w is the local relative velocity [23]. With $\mathrm{Pr} \approx 0.70$ for cold air, $r \approx 0.89$, and at the design rotor inlet relative velocity the recovery term adds back around 58 K, so the blade surfaces equilibrate near −30 °C rather than the static −88 °C.

Two follow up checks close the question. First, the blade thermal time constant. Treating the thin blade as a lumped capacitance with the film coefficient from a turbulent flat plate correlation evaluated at the blade chord, the time constant $\tau_{th} = \rho_b\,(t/2)\,c_{p,b}/h$ comes out below one second, so the blades reach their steady temperature within seconds of the start of a run and there is no transient grace period to exploit. Second, dimensional stability: the thermal strain at 50 K below ambient with the resin's expansion coefficient of order 10^{-4} per K is about 0.6%, which over the 4 mm blade height is some 25 µm, small against the running clearances. The thermal design conclusion is therefore that the geometry is unaffected but the material toughness at $-30\,°\mathrm{C}$ is the binding concern, which is what motivated the truncated trailing edge flat of the previous section.

3.2.9 Structural and Rotordynamic Analysis

The rotor was checked with static FEA at the design speed: blade root bending under the combined centrifugal and gas loads, blade deflection, shroud hoop stress and disc bore stress. The blade bending safety factor is 39 on the Tough 2000 material. All FEA was run in Fusion 360 static studies. For linear elastic statics the solver brand is irrelevant, and each result was cross checked against a hand calculation of the matching idealised case, which is the verification that actually matters.

The hand calculations are the classical rotating component results. The peak hoop stress of a solid rotating disc bounds the disc and bore stress,

$$\sigma_{\theta,max} = \frac{3+\nu}{8}\,\rho_b\,\omega^2 r^2 \tag{73}$$

the centrifugal pull at the blade root is the integrated body load of the blade above it,

$$\sigma_{root} = \frac{\rho_b\,\omega^2}{2}\left(r_{tip}^2 - r_{root}^2\right) \tag{74}$$

and the shroud ring carries its own centrifugal load as a free hoop,

$$\sigma_{hoop} = \rho_b\,u_{shroud}^2 \tag{75}$$

where ρ_b is the blade material density and u_{shroud} the shroud rim speed. The gas bending load on each blade adds a cantilever bending stress $\sigma_b = Mc/I$ at the root, with the moment from the blade force of the velocity triangles. Each FEA peak stress was required to agree with its idealised counterpart to the level the idealisation allows, which catches unit, constraint and load errors that mesh refinement cannot.

For the shaft, the finite element rotordynamics of fig. 20 is cross checked by the overhung rotor estimate. Treating the shaft as a cantilever of bending stiffness EI with the disk mass m at radius L from the bearing,

$$\omega_1 \approx \sqrt{\frac{3EI}{mL^3}} \tag{76}$$

which lands in the same regime as the finite element first bending pair and confirms the order of the margin.

For rotordynamics, the shaft is a 10 mm solid aluminium stub. A finite element rotor model with the turbine disk overhung and the measured disk mass and inertias places the first bending natural frequency at standstill around 99,000 RPM, and the Campbell diagram in fig. 20 shows the margin over the full operating range. The first synchronous crossing sits far above the 20,000 RPM design speed, with a margin of better than 3.5. The shaft connects to the pump through a jaw coupler with Double-D 7 mm flats, sized with a compressive safety factor of around 10 on the printed hub.

Figure 19: Rotor static FEA at 20 000 rpm: blade root bending, shroud hoop stress and disc bore stress, with the resulting safety factors.

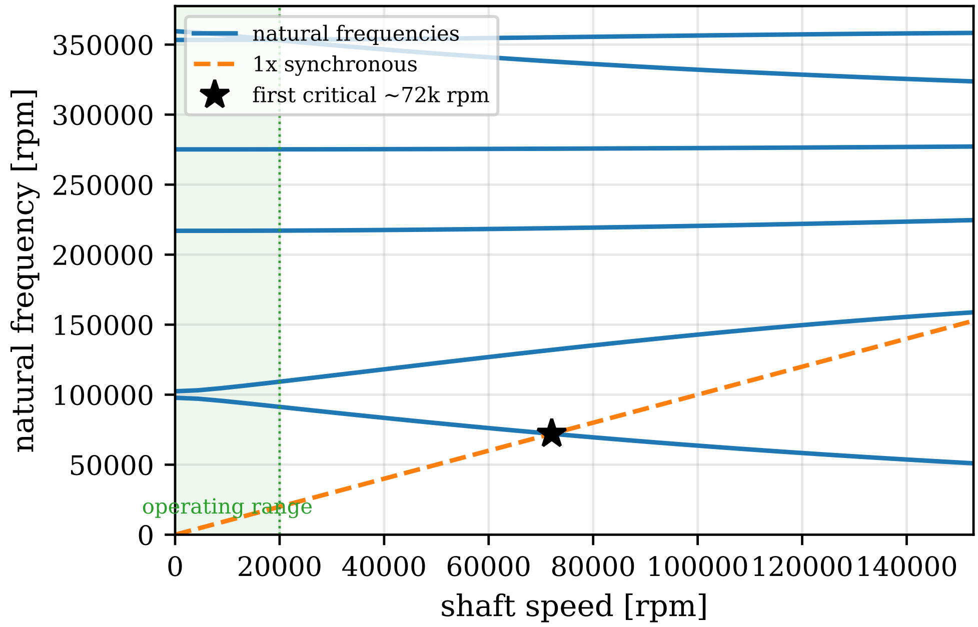


Figure 20: Campbell diagram of the turbine shaft from a finite element rotor model (10 mm solid aluminium shaft, turbine disk overhung, ball bearing stiffness; the coupler side disk is omitted). The first bending pair starts near 99,000 RPM at standstill and the backward whirl branch crosses the synchronous line at around 72,000 RPM, leaving a margin of more than 3.5 over the operating range.

3.2.10 Manufacturing

The rotor and the stator nozzle block were printed by SLA in Tough 2000 resin, washed in IPA, UV cured, and the mating faces sanded flat. Printed resin turbomachinery at this power level is published practice rather than an improvisation, as Weiss cites both an experimental kW scale SLA resin air turbine and SLA diffuser vanes in operation [24].

4 Experimental Setup

The experimental setup was designed for three specific configurations: Electric motor driven pump testing, Isolated turbine coupled to a hysteris brake, and integrated turbine and pump testing.

4.1 Test Configurations

(a) Motor driven pump

(b) Turbine against brake

(c) Coupled turbopump

Figure 21: The three test configurations. In each case the TM306 torquemeter sits between the driving component and the load, so torque is measured at the shaft.

In all three configurations, there is a driving component, the torquemter and a load. fig. 21a shows the electric motor driven pump testing configuration. The electric motor test is used to gather most of the data because the motor is more controllable, repeatable and safer. In this configuration. The second configuration in fig. 21b connects the turbine to a hysteris brake to test the turbine in isolation. Due to time constraints, this configuration was not tested. The final configuration in fig. 21c connects the turbine and pump together to test the performance of the integrated system.

4.2 Mechanical Layout and Shaft Train

At the design 20,000 RPM for the rig, shaft alignment, rigidity and maintainability were the main drivers. At 20,000 RPM, available shaft couplers had barely any tolerance for misalignment. The Magtrol MIC-5-2470 couplers had a maximum

0.8mm axial, 0.36mm radial and0.7° angular misalignment tolerance at 35,000 RPM. Shaft couplers had to be used to connect rotary components to the transducers, a requirement that doesn't have to be satisfied by turbopump units as the shaft can be machined as a single piece.
The central component of each configuration in fig. 21 is the Magtrol's TM306 inline torquemeter as it allows the stated ¡0.1% torque measurement accuracy. As the torquemeter was designed to be mounted onto a base plate, a base plate design was chosen instead of an integrated assembly like a normal turbopump.
The base plate is a 20 mm plywood with a laminate on both sides to be semi-waterproof, to have a flat surface for mounting, rigid but also cheap. M6 and M8 Countersunk holes were drilled into the base plate with a CNC router to allow for precise mounting of the components.

Figure 22: The assembled test rig with the major components labelled: motor, couplers, torquemeter, pump, turbine and the throttle valves on the CNC drilled baseplate.

4.2.1 Shaft alignment

The reason the shafts had to be aligned was because flexible shaft couplers were used. These tolerate some misalignment at high speeds, an essential requirement for a test rig where two or more shaft have to be mechanically connected.
The alignment of each component to the torquemeter had many challenges. Mounting magnetic dial indicators onto the shafts was not possible due to three reasons. The shafts were made of austenitic stainless steels so they were non-magnetic, the shafts were short and had a low diameter (10-15mm diameter, with maximum stub length of 20mm). A short stub length is highly desirable in rotordyanmics to increase rigidty and increase the first critical speed.
The shaft couplers were also single piece, so the shaft couplers which do not have a edge true to the shaft axis. This meant that the shaft couplers could not be installed after the shafts were aligned, as the couplers would have to be installed before the shafts were aligned. The space the shaft couplers took up meant it was even more difficult to use the shaft stubs for alignment.
It was decided to use the flat faces of the torquemeter base as the reference for alignment. By designing the pump and turbine mounts to have a similar rectangular base and assuming that each shaft was true to the rectangular base, alignment was finally possible. Angular alignment was achieved by 3D printing a spacer with two parallel edges for the pump or turbine to sit flat against onto the torquemeter. Then, radial misalignment was achieved by designing the face parallel to the shaft to be the parallel to the torquemeter face. A straight edge could then be used as a guide to align the shaft while tightening the mounting bolts. This procedure is shown in fig. 23.

Figure 23: Shaft alignment procedure using the flat faces of the component mounts and precision square blocks as references.

As the TM306 torquemeter's shaft with the base plate was machined to be preciscely 80.00 mm from the base plate, this same height was built into the pump and turbine mounts to achieve radial aligment in the vertical direction.

4.3 Water Circuit

Figure 24: P&ID of the water circuit. The tank feeds the supply pump, then the inlet throttle valve, the experimental pump, the outlet throttle valve and the return line. Pressure is measured at the supply ($\mathrm{pt_{pump}}$), the pump inlet ($\mathrm{pt_{in}}$) and the pump outlet ($\mathrm{pt_{out}}$), and flow in the inlet line (fm0).

The water circuit is as follows: A water tank is pumped by the supply pump into an inlet throttle valve then into the experimental pump inlet. The pump outlet is then throttled by an outlet throttle valve before being returned to the water tank. Pressure sensors are placed at the inlet and outlet of the pump to measure the head rise across the pump. Flow rate is measured by a flow meter placed in the inlet line before the inlet throttle valve instead of the return line due to space constraints placed by the blast shield.

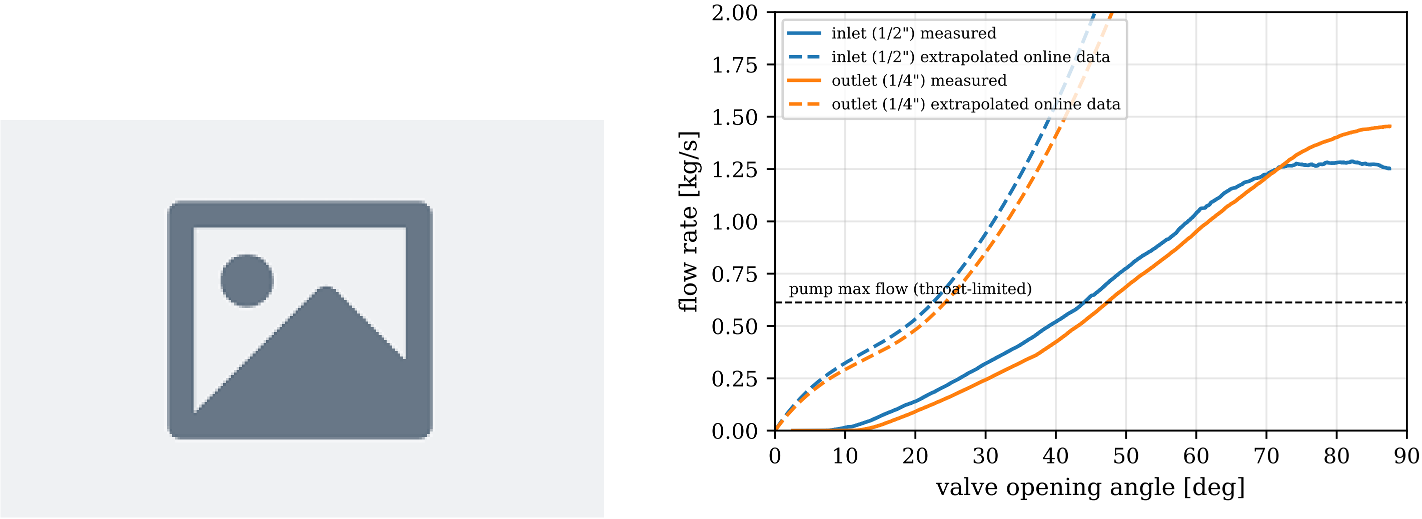
4.3.1 Inlet and Outlet Throttle Valves

The inlet throttle valve allows a reduction in pump inlet pressure to characterise the cavitation behaviour of the pump. The outlet throttle valve allows for a reduction in flow rate to characterise the head and cavitation behaviour of the pump at different flow rates.

The selection of the size of the inlet and outlet valves are crucial to the quality of the data. Ball valves were chosen as there were spares available, and they could be throttled in a controllable and repeatable manner by the angle of the valve handle. Custom 3D printed housings were designed to attach a servo motor to the valve handle in direct drive to allow for precise, remote and repeatable control of each valves.
Valve capacity is expressed throughout with the flow coefficient K_v, defined by

$$Q\,[\mathrm{m^3/h}] = K_v \sqrt{\frac{\Delta p\,[\mathrm{bar}]}{SG}} \tag{77}$$

where SG is the specific gravity of the fluid, 1 for water. Measuring the flow rate and the pressure drop across a valve at a known opening angle therefore measures its K_v directly, which is how the characterisation below was done.
The main requirement for both valves were so that the range of required flow rates and pressure drop through each valve corresponded to a large range in valve opening angles. For the inlet valve, fully open must have a large margin to the cavitation limit of the pump, ideally, cavitation would start at around 50% valve opening and could be further reduced until the valve is closed. For the outlet valve, fully open must be at least 1.5 times the cut-off flow rate of the pump.
This requirement is further driven by the $1-3°$ resolution of the servo motors available in the department. Data for Kv against opening angle were extrapolated from online data from 1/8" in to 3/4" in were studied. However, as the data was extrapolated from data of ball valves with larger diameters, they had to be experimentally validated. The flow through a ball valve is also highly dependent on inlet and oulet conditions such as contraction or expansion of pipe diameter at the ports [?]. The size of my pipes were limited to what I could find in spare drawers, adding to the uncertainty.



(a) Servo driven throttle valve assembly

(b) Extrapolated against measured valve capacity

Figure 25: Throttle valve characterisation. The extrapolated online ball valve data (dashed) overpredicts the measured flow capacity (solid) for both valves at their respective design pressure drops. The horizontal line is the throat limited maximum flow of the pump.

fig. 25b shows the experimentally measured Kv against opening angle values were lower than expected. However, this worked in my favour as it actually increased the range of useful valve opening angles.
The location chosen for the experiment required a safe test area and a control room separated to allow for the failure of the test rig without risk to the operator. For this reason, the experiment was conducted within the Imperial College Hypersonics Lab. The test room contains equipment that had to be protected, so a blast shield was designed and set up around the turbomachinery stand to protect the equipment from the possibility of mechanical failure and fragmentation.

4.4 Air supply

Originally, a 100 bar compressed air supply was designed and parts readied to be used to drive the turbine. Unfortunately due to time constraints, the shop air supply within the hypersonics lab was used instead. Details of the 100 bar compressed air supply can be found in . The shop air which was limited to 7 barg was connected to partially assembled parts of original feed system. The shop air was connected to a GRT20s-280N100-SSKN-L 0-280 to 0-100 bar regulator with a Kv of 1.5. This regulator fed into the the main 1" valve, teed into a 1/2" vent valve and finally into the turbine inlet.
Both valves were Swagelok pneumatically actuated spring return ball valves. The main valve was the SS-65TS16-35C with a Cv of 40 and the vent valve was SS-45S8-33C with a Cv of 12. These valves were driven by 2 5/2 solenoid valves connected to the shop air supply.
Due to droop in the shop air regulator, maximum turbine inlet pressure at max flow was measured to be 3.1 bara. There was no budget for an air flowmeter.
The supply droop deserves a proper explanation because it directly sets the coupled test conditions in section 5. The shop air behaves as a regulated blowdown with a finite supply. When the turbine valve opens, the regulator cannot hold its set

pressure at the demanded flow, so the stator pressure sags well below the static setting, and the achievable turbine inlet pressure under flow is roughly 38% of the design supply. This is a property of the rig, not of the turbine, but it has to be characterised because the turbine match point depends on it.

To verify the original 100 bar feed system before it was descoped, a transient flow network simulation was written from scratch. The model solves the tank blowdown as a system of differential equations, with each component represented by its flow coefficient, tube friction from roughness and bends, a regulator model holding its set pressure within a control band, and the choked stator as the terminating element. This was admittedly an overkill way to answer the question of whether two 3.8 litre tanks held enough gas for a full test, but it gave the test durations and the pressure transients at every node of the feed system, and the same component models were reused to size the valve Kv requirements. The simulation code is part of the project's analysis repository.

4.5 Instrumentation and Data Acquisition

For cost and time reasons, a custom data acquisition and control system was designed to allow for remote operation of the test rig.

4.5.1 Main Architecture

Figure 26: Data acquisition architecture. Sensors and the DSP6001 analog outputs feed the MCP3208 ADC, read by an ESP32-S3 over SPI, streamed to the Raspberry Pi 5 over UART, logged to HDF5 and forwarded to the control room GUI over the network.

For safety reasons, the electronics were enclosed in a plastic box placed in the test room, controlled by a LAN cable fed under the wall between the test room and the control room into my laptop.

The main main computer inside this box was a Raspberry Pi 5 running a backend written in Python. The Raspberry Pi communicated with 3 ESP32S3 microcontrollers over a UART serial connection at 921600 baud. One ESP32S3 was used as the actuator controller, one was used as a ADC and flowmeter DAQ, and one was used as a thermocouple DAQ. Breadboards would require too many wires to maintain and unstable, while there was not enough time to make custom PCBs. Soldered perfboard was used to connect the components. The Raspberry Pi communicates through the LAN cable through UDP to a laptop in the control room, allowing for live data readout and control of the test rig with a GUI written in Python and Qt.

Figure 27: The control room GUI during a test: live channel readouts, strip charts of pressure, torque and speed, valve angle controls, and the test sequencing and logging controls. Every quantity shown is logged to HDF5 at the full 50 Hz loop rate, so the analysis of section 5 works from exactly what the operator saw.

4.5.2 ADC Measurements

Pressure, torque, RPM, voltage and current were measured with a MCP3208 8 channel 12-bit ADC connected to the ESP32S3. The MCP3208 was powered by a LT3042 LDO at 5.11V to reduce noise. This 5.11V was also used to power the pressure sensors. As the pressure transducers were ratiometric, both the ADC and the pressure transducers track bacck to the same voltage reference, reducing drift and noise.

The custom backend could only send UART signals back and forth to the ESPs at a 50 Hz loop. The MCP3208 was capable of sampling up to 100 ksps, so the ESP32 was programmed to collect 32 samples and average them per channel before sending the averaged data back to the Raspberry Pi. High frequency data couldn't be collected due to the UART connection being the bottleneck, but this averaging allowed for measurement of voltage at a resolution higher than 12-bits. For uncorrelated noise the averaging reduces the standard deviation of each logged point by

$$\sigma_{avg} = \frac{\sigma}{\sqrt{N}} = \frac{\sigma}{\sqrt{32}} \tag{78}$$

which is a factor of about 5.7, and is why the per bin scatter in the results is small.

The ratiometric arrangement deserves one line of algebra, because it deletes a systematic error term by design. The transducer output is proportional to its excitation voltage, $V_s = k\,p\,V_{exc}$, and the ADC reports the ratio of its input to its reference, $D = V_s/V_{ref}$. With both supplied from the same LT3042 rail, $V_{exc} = V_{ref}$, so

$$D = \frac{k\,p\,V_{exc}}{V_{ref}} = k\,p \tag{79}$$

and any drift of the shared rail cancels to first order, leaving only the transducer's own nonlinearity and the reference noise.

Originally, a separate computer running Magtrol's M-Test software was planned to be used to read the torque and RPM from the DSP6001. However, by connecting the DSP6001's analog output for torque and RPM to the MCP3208, the torque and RPM data could be integrated into the same DAQ system as the pressure sensors, allowing for time-exact data synchronisation and simpler data collection.

To measure the torque and RPM, the TM306 torquemeter was connected to the Magtrol DSP6001 dynamometer controller. The DSP6001 outputs a -10V to 10V analog signal for torque, and 0 to 10V for RPM. The DSP6001 was set to output 1V per 0.25 Nm and 1V per 5000 RPM. Because the MCP3208 could only measure positive voltages, an OPA2134 op-amp was used to shift the expected -5V to 5V torque signal to 0V to 5V. As the op-amp was not rail-to-rail, a buck converter was used to generate a -9V and 9V supply for the op-amp.

The motor's voltage and current were also measured by the ADC. The 24V supply to the motor ESC was measured with a voltage divider, and the current was measured with a WCS1600 DC hall effect current sensor.

Figure 28: ADC and signal conditioning schematic: the LT3042 reference powering both the MCP3208 and the ratiometric pressure transducers, and the op-amp level shift for the torque signal.

4.5.3 Other sensors

K-Type thermocouples although are less accurate for such a small temperature range were chosen for their cost. The thermocouples were read by an Adafruit MAX31855 thermocouple amplifier breakout board with integrated voltage regulator connected to the ESP32S3.
A E-WFS-GE turbine flowmeter with a range of 2 to 45 L/min with a cited accuracy of ±10% was used. To calibrate the flowmeter, a bucket and stopwatch was used to compare the flowmeter to the integrated flow rate over a period of time. The flowmeter was found to be accurate to ±3% [**?**]. The turbine generates pulses based on its rotation rate. The ESP32 measures the interval between these pulses to calculate the flow rate.

4.5.4 Actuator Control

This ESP32 was responsible for controlling the inlet and outlet throttle valve angles,

4.5.5 Cavitation Imaging

The transparent front casing and diffuser turn the pump into its own instrument, but only if the bubbles can actually be seen on camera. Cavitation bubbles are transparent; what a camera records is light scattered at the bubble interfaces, so the imaging arrangement matters as much as the casing. The casing was lit obliquely with a high intensity lamp placed off the camera axis, so that bubble interfaces scatter light toward the camera while the bulk water stays dark, and the polished resin surface was angled to keep the specular reflection of the lamp out of frame. Video was recorded through the polycarbonate shield with a consumer camera.
Two limitations of this arrangement should be stated. The frame rate is orders of magnitude below the blade passing frequency, so each frame is a time averaged record of the cavity envelope, not a snapshot of individual bubble dynamics, and the consumer camera's rolling shutter means different image rows sample different instants, geometrically distorting fast moving structures. Individual bubbles are still resolved in some frames where the motion is slow enough. These limits are acceptable for what the video is used for in section 5, identifying where cavitation occurs, when it onsets, and when the throat clears, all of which are envelope level questions. Bubble tracking or cavity dynamics would need a global shutter camera with a synchronised strobe, which is noted as future work.

Figure 29: The cavitation imaging arrangement: oblique illumination of the transparent casing with the camera viewing through the blast shield.

4.5.6 Instrument Uncertainty

table 5 lists every measurement channel with its range and systematic uncertainty. These values feed directly into the uncertainty analysis of section 5, where they are combined with the per point random scatter. The torque chain is the best characterised: the TM306 itself is rated to better than 0.1% of its 5 Nm range, the DSP6001 analog output adds 0.05%, and the conditioning chain was designed so the dominant systematic terms cancel, as described in the previous section. Where calibration certificates were not available, cheap cross checks bound the errors instead. The measured efficiency landing close to the design estimate validates the torque scale, the shutoff head scaling with speed squared through the origin bounds the pressure transducer gain error, and the repeatability of the non dimensional head coefficient across five speeds bounds the random error.

Table 5: Measurement channels, ranges and systematic uncertainties.

Quantity	Sensor	Range	Output	Uncertainty	Source
Pressure	TODO	TODO	TODO	TODO	TODO
Flow rate	TODO	TODO	TODO	TODO	TODO
Torque	TODO	TODO	TODO	TODO	TODO
Shaft speed	TODO	TODO	TODO	TODO	TODO
Temperature	TODO	TODO	TODO	TODO	TODO

4.6 Test Envelope and Limitations

The achievable test envelope was set by the drive hardware on the pump side and by the air supply on the turbine side. The electric motor had no speed control loop, only stepped throttle commands to the ESC, and the combination of the ESC current limit and motor torque cap meant pump testing topped out at around 7,500 RPM. On the turbine side, the shop air pressure and its droop under flow settled the coupled shaft at around 5400 RPM. Conditions above these speeds were not surveyed due to these experimental limitations, and the extrapolation of the measured behaviour towards the design point is handled analytically in section 5.

Safety drove three features of the test arrangement that are visible in the data taking: all testing was operated from a separate control room with nobody in the test cell, the rotating assembly was fully enclosed by a blast shield, and a hardware safety relay cuts all actuation power on an overspeed or emergency stop. The shield and the protection system were not bought off a shelf, they were sized by the calculations of the next section.

4.7 Containment and Protection System Design

Experimental turbomachinery has to be designed on the assumption that any rotating component or any non commercial pressure part can fail. The containment hardware was therefore sized from first principles. A note on the operating pressure: these calculations were performed for the original 100 bar stored air system that was later descoped to the 7 bar shop air supply, and the hardware they sized was retained, so the as-tested system carries safety factors far beyond what the 7 bar regime requires.

4.7.1 Fragment Energy and Blast Shield Sizing

For the failure of a pressure holding component, the fragment velocity is estimated with the method of Baker [25], developed from simulations of a vessel breaking into sectional fragments and matched to experiment within 10%. The non-dimensional burst pressure is

$$\bar{P} = \frac{(P_0 - P_e)\, V_0}{m_c\, \gamma R T_0} \tag{80}$$

where P_0 is the vessel pressure at failure (the full 100 bar, assuming both the regulator and the burst disc have failed), P_e is atmospheric pressure, V_0 the vessel volume and m_c the casing mass. The worst case line of Baker's velocity chart is

$$\log_{10} \bar{u} = 0.6 \log_{10} \bar{P} + 0.15, \qquad \bar{u} = \frac{u}{\sqrt{K \gamma R T_0}} \tag{81}$$

with $K = 1$ for evenly distributed fragment mass. The only non commercial pressure holding component is the stator, and this procedure gives a fragment velocity of 72 m/s. Assuming the stator splits into two 0.1 kg fragments,

$$E = \tfrac{1}{2} m u^2 = \tfrac{1}{2} \times 0.1 \times 72^2 = 261 \text{ J} \tag{82}$$

For the rotating assembly, the bounding energy is the full rotational kinetic energy of the blisk at design speed,

$$E = \tfrac{1}{2} I \omega^2 = \tfrac{1}{2} \times 6 \times 10^{-5} \times 2094^2 = 131 \text{ J} \tag{83}$$

The main line of defence is a 20 mm Dyneema (UHMWPE composite) ballistic panel on an aluminium extrusion frame. Its capacity is expressed as a specific energy absorption,

$$SEA = \frac{E}{m}, \qquad E_{max} = SEA \cdot A\, t\, \rho \tag{84}$$

where A is the projected area of the impacting fragment, t the panel thickness and ρ the panel density. The SEA is derived from published v_{50} ballistic limits for fragment simulating projectiles against 20 mm panels [26]: a 12.7 mm FSP gives 1.84 MJ/kg and a 20 mm FSP gives 1.68 MJ/kg, the latter agreeing with independent flat projectile data. Using the conservative value, the panel can absorb 163 kJ over the stator's projected area and 32.6 kJ over the blisk's, against the fragment energies of eqs. (82) and (83). The containment margins are roughly 600 and 250.

A 15 mm polycarbonate shield additionally separates the turbomachinery from the dynamometer. Its capacity is estimated with the shear plug model [26], where the projectile shears a plug of perimeter P_{im} out of the sheet:

$$E = \int_0^t P_{im}\, \tau\, (t - x)\, dx = \tfrac{1}{2} P_{im}\, \tau\, t^2 \tag{85}$$

with $\tau = 60$ MPa for polycarbonate, conservative because the material strain hardens at impact rates. This gives 2.7 kJ for the stator perimeter and 4.1 kJ for the blisk perimeter, safety factors of 10 and 30 on their own.

4.7.2 Bounding Shaft Torque

The rig torque rating was checked against an upper bound from the Euler equation, taking the maximum mass flow, full tangential jet velocity at entry and perfect counter rotation at exit:

$$\tau_{max} = \dot{m}\, r\, (c_{u,in} - c_{u,out}) = 0.034 \times 0.05 \times (469 + 469) = 1.6 \text{ Nm} \tag{86}$$

This is an unachievable worst case and still sits well below the 3 Nm rating of the shaft train.

4.7.3 Protection System Electronics

The protection logic that the next section's transient analysis relies on is implemented in hardware, not software. The core is a PILZ PNOZ X3 safety relay wired as a dual channel emergency stop chain: the control room E-stop button and the overspeed trip both open the chain, and the relay then cuts the supply to every actuator solenoid through its force guided contacts. The safety relay is a certified device whose failure modes are themselves controlled, which is the reason for using one instead of a software interlock through the Raspberry Pi: the software stack is involved in detecting the overspeed, but once the trip fires, no software is in the loop between the relay and the valves. The pneumatic actuators are spring return, chosen so that the de-energised state is the safe state: on any loss of actuation power, including a tripped relay, a cut LAN cable or a power failure, the main air valve drives closed and the vent valve opens the manifold to atmosphere. The 20 ms figure used in the runaway analysis below is the relay contact response; the valve travel times dominate the rest of the shutdown timeline.

The electrical system itself is low risk by construction: everything in the test cell runs at 24 V DC or below from a single supply, the electronics live in an enclosed plastic box protecting them from spray, and the only conductor crossing the test cell boundary is the LAN cable to the control room, which carries no power. The motor ESC and its 24 V supply are the highest energy electrical components, and they are inside the same enclosure boundary with the rotating hardware, behind the shield.

4.7.4 Overspeed Runaway Transient

The protection system has to act faster than the worst credible overspeed. With a broken shaft leaving only the turbine, blisk and bearings, the angular acceleration under the maximum torque at the 17,000 RPM redline is

$$\alpha = \frac{\tau}{I} = \frac{0.9}{2.74 \times 10^{-4}} = 3285 \text{ rad/s}^2 \approx 31{,}300 \text{ RPM/s} \tag{87}$$

After the trip, the manifold between the closed valve and the stator continues to blow down through the turbine. In choked flow the mass flow is proportional to the remaining manifold mass, an exponential decay with time constant

$$\frac{dm}{dt} = -\dot{m}_0 \frac{m}{m_0} \quad \Rightarrow \quad \tau_{bd} = \frac{m_0}{\dot{m}_0} = 42 \text{ ms} \tag{88}$$

for the measured manifold volume. The protection sequence is: overspeed detected at $t = 0$, all actuation power cut by the hardware relay at 20 ms, main valve fully closed at 170 ms, manifold vented by around 200 ms. Holding the maximum torque over the full 200 ms overestimates the excursion and still bounds it at 6,260 RPM, so an overspeed tripped at 17,000 RPM peaks below 23,300 RPM, briefly exceeding the 20,000 RPM dynamometer rating within the manufacturer's stated tolerance for transient overuse. Cavitation testing was further speed limited so that the same worst case stays below 19,000 RPM. The venting noise and reaction force calculations that complete the safety case are in section B.

5 Results and Discussion

5.1 Experimental Campaign

The experimental campaign was conducted in three phases: commissioning, electric motor driven pump testing, then integrated turbine and pump testing. The standalone turbine testing was not conducted due to time constraints. This means the turbine is assessed through its models and through the coupled test, and a standalone brake characterisation is the first item of future work.

Within each run the shaft speed was not perfectly constant, so the data points were adjusted to the run median speed using the affinity laws before binning. The adjustment is small, validated to around 0.2% on the head coefficient, and it is what allows each run to be treated as a single speed curve.

All error bars in this chapter follow the same simple scheme. The systematic uncertainty of each instrument comes from table 5, the random uncertainty is the standard error of the median within each bin, and the two are combined in quadrature at roughly 95% confidence, $U_Y = \sqrt{B_Y^2 + (2P_Y)^2}$. Propagation into derived quantities like efficiency and the non dimensional coefficients is done by linear propagation. One design decision from section 4 matters here: because the pressure transducers and the ADC share the same reference voltage, excitation and gain drift cancel to first order and do not appear as a systematic term. Where error bars are not visible they are smaller than the markers.

5.2 Pump: Head–Flow Characteristics

There are two main curves that are used to evaluate pump performance: the head-flow curve and the efficiency curve. The head-flow curve shows the achievable head at different flow rates and will be discussed first.

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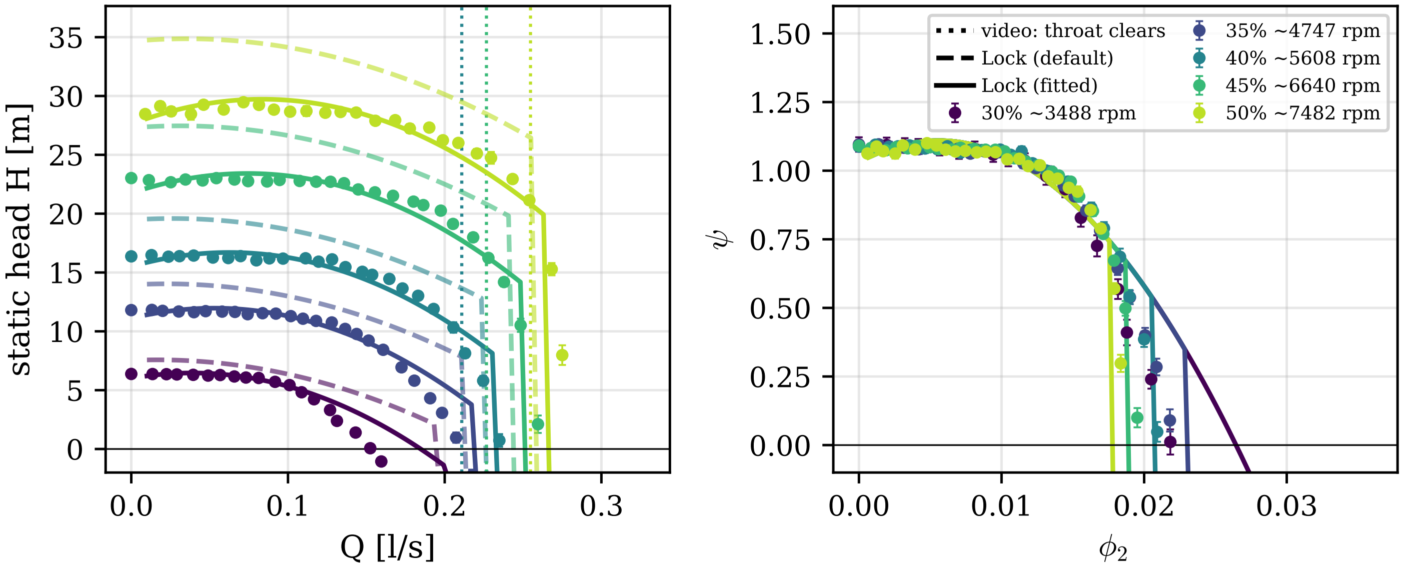

Figure 30: Measured head against flow rate across five speeds (left) and the same data collapsed to head and flow coefficients (right). Dashed lines are Lock's model with literature constants, solid lines are the bounded re-fit with the as-built throat. Vertical dashed lines mark where the synchronised video shows the diffuser throat cavitation clearing. Error bars are systematic and random combined in quadrature at 95% confidence, smaller than markers where not visible.

fig. 30 Shows the actual achieved head rise across flow rates at the different RPMs that were tested. Overlayed are solid and dashed lines. Both are theoretical head-flow curves generated from eq. (25). The dashed line is with exactly the parameters from Lock's paper, and the solid line is with the pre-rotation K increased from 0.17 to 0.53. The dashed vertical dashed line shows the point at a synchronized video camera detected cavitation within the diffuser through the transparent diffuser wall. It can be seen that the original parameters overpredict the head rise, and there are many mechanisms that could be causing this.

The non dimensional view in the right panel carries the headline numbers. The five speeds collapse onto a single curve at a head coefficient of $\psi = 1.10 \pm 0.05$ in the operating region, against a design value of 1.065, so the pump delivers its design head with around 3% to spare. The collapse itself is the experimental proof of dynamic similarity, and it is what licenses the affinity adjustment within runs and the extrapolation to higher speeds later in this chapter. The diverging tails at high flow coefficient are a rig artifact rather than a breakdown of similarity, because the fixed feed pressure means the maximum achievable flow coefficient falls as speed rises.

The fitted constants need to be defended rather than just stated, because a fitted model proves nothing if the constants are unphysical. The bounded re-fit with the as-built throat diameter fixed at 3.0 mm gives $K_1 = 0.53$ and a diffuser loss coefficient of 0.42, both interior to their physical bounds. The excess of K_1 over Lock's own 0.15 to 0.20 study has named candidates that are all in Lock's paper: the deliberate prerotation designed in through $d_1 = 1.1d_0$, the rounding of the printed blade leading edges which Lock notes reduces effective head, and the optimism of his effective radius assumption. The diffuser loss of 0.42 against the clean pipe entry value of 0.194 is consistent with the entry separation mechanism Lock describes at the diffuser mouth. Alternative explanations can be ruled out: an error in the torque or pressure scales would shift the dimensional curves but not change the shape of the droop, and a slip type explanation would move the intercept of the non dimensional curve rather than its slope.

The most important feature of fig. 30 is the high flow cutoff, and here the transparent casing converts a fit into a validation. On every head flow ramp the video shows cavitation in the diffuser throat clearing as the flow falls, frame matched to the data at three flow rates across three runs. With the true throat diameter of 3.0 mm in the model, Lock's vapour pressure cutoff lands on all three video clearing points to within roughly 10%, and on the measured runout. It is worth being honest about the history here: with the nominal cutwater diameter of 3.8 mm the model overpredicted the cutoff badly, and it was the systematic disagreement with the video that prompted the CAD section check which found the true minimum area. The model diagnosed a wrong geometry input, which is a stronger validation than agreement on the first try. The remaining discrepancies are also informative. The measured cutoffs sit 5 to 10% later than predicted, consistent with a small residual flow contraction, and the head begins rolling off slightly before the cavitation clears, which cavitation cannot explain and which points to the same diffuser entry separation responsible for the elevated loss coefficient. Collector losses growing with flow at low specific speed is independently reported, as Khan's volute head loss curves blow up at overload in exactly this way [27].

The conclusion of this section is then a mechanism, not a fit: the maximum flow of this pump is limited by cavitation in the diffuser throat, established independently by the model and the video.

5.3 Pump: Efficiency and Churning

The overall efficiency is computed from directly measured quantities only,

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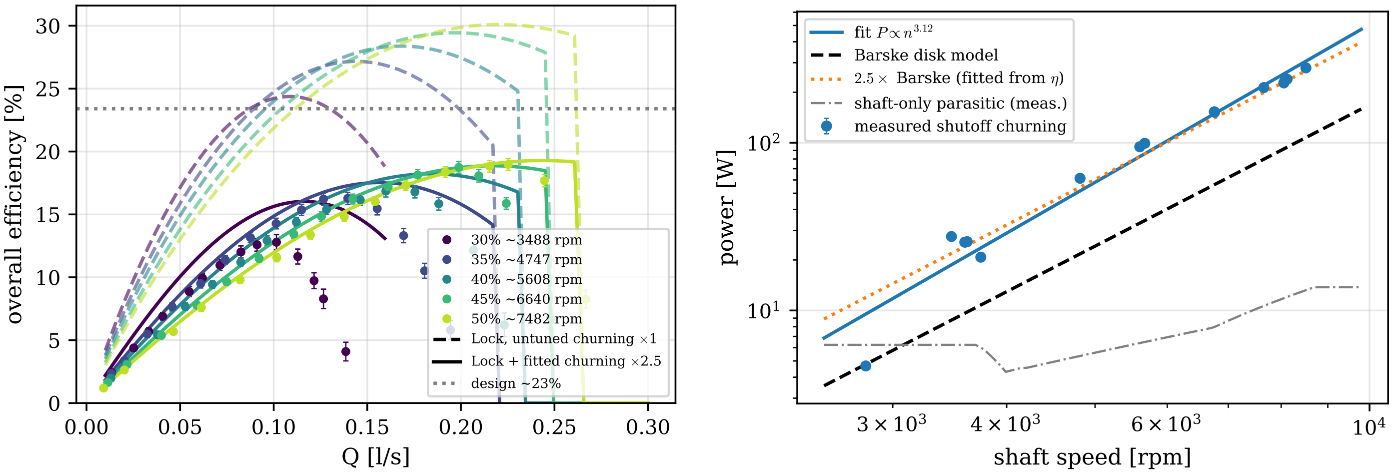
$$\eta = \frac{P_{hyd}}{P_{shaft}} = \frac{\rho g Q H}{\tau \omega} \tag{89}$$

with the torque measured at the shaft between the drive and the pump, so the drive losses never enter.
The efficiency curve generated by measuring shaft power, subtracting mechanical losses can be shown in the following graph: The right graph shows the predicted efficiency curve from pure theory, and the right graph shows the curve but with adjusted disc friction. To explain the disc friction adjustment, the pump data at zero flow where all the shaft power was only going into hydraulic losses:



(a) Overall efficiency against flow rate

(b) Shutoff churning power against speed

Figure 31: Pump efficiency and the churning measurement. In (a) the dashed curves use Barske's textbook disc friction and overpredict the efficiency, while the solid curves apply the measured churning multiplier. In (b) the churning power is measured directly at shutoff, where shaft power minus the measured shaft-only parasitic is all churning; the full campaign of 15 shutoff points is used, not just the five canonical runs. Error bars as in fig. 30.

fig. 31b shows that the measured disc friction losses are about 2.5 times higher than barske's formula for disc friction. Adjusting the disc friction by this factor leads to a much better fit of the efficiency curve, as can be seen in fig. 31a.
The way this multiplier was measured matters, because it changes its status from a tuning factor to a measurement. At shutoff the pump moves no flow, so after subtracting the measured shaft-only parasitic power, everything the shaft delivers goes into churning the flooded casing. Across 15 shutoff points from 2,800 to 8,500 RPM the measured churning sits at 2.5 to 2.9 times the Barske disc correlation, with a mean of 2.54, and follows a power law of $n^{3.1}$, between Barske's exponent of 2.8 and the cubic forms in Gülich. The factor of 2.5 used in the efficiency model was originally fitted to the efficiency curves, so its agreement with the shutoff measurement is corroboration by data that did not produce it. The points below about 4,800 RPM scatter heavily and were excluded, as the torque there sits at the noise floor of the measurement chain.
The peak measured efficiency is $(19 \pm 2)\,\%$ against a design estimate of $23.4\,\%$. Where the power actually goes at the best efficiency point is summarised in fig. 32. Churning dominates everything: at the 7,500 RPM best efficiency point roughly two thirds of the shaft power goes into churning the flooded casing, mechanical losses take a few percent, the internal flow losses take around a tenth, and only about a fifth of the shaft power leaves as useful hydraulic power. The attribution to the open impeller architecture is the point, as the whole casing is flooded and churned while only one sector does useful pumping. This is the price of the Barske architecture at low Reynolds number, not a defect of this particular build.

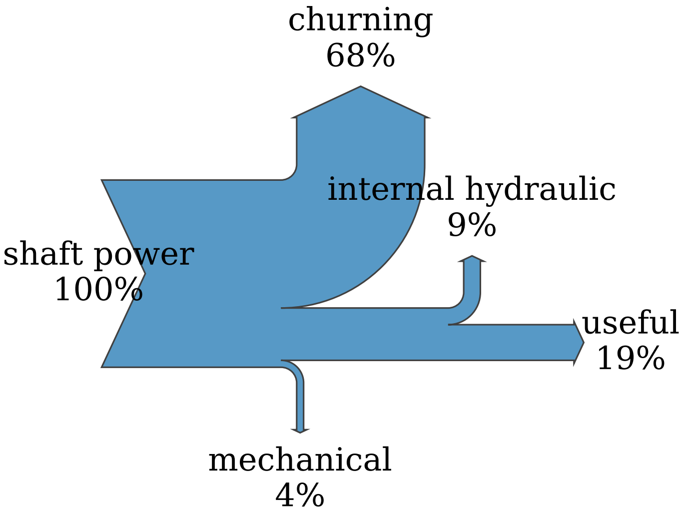


Figure 32: Shaft power budget at the measured best efficiency point of the 7,500 RPM run. Useful hydraulic power is measured, churning uses the shutoff-measured multiplier at this speed, mechanical losses are the measured shaft-only parasitic, and the remainder is internal hydraulic loss.

The natural question is what this efficiency becomes at the design speed, and the honest answer comes from the data rather than an assumption. Each run sits at a different Reynolds number, so the five runs themselves measure how the losses scale. fig. 33 plots the peak efficiency of each run against speed and fits the loss ratio model $(1-\eta) \propto Re^{-a}$, which is the standard form for friction dominated loss scaling. The fitted exponent is $a = 0.098 \pm 0.010$, essentially the textbook Moody value of 0.1, and extrapolating along the fit gives an efficiency of 26.4% at 20 000 rpm with a band of 25.7 to 27.1%. The exponent is measured from this machine, not assumed, which is the difference between an extrapolation and a guess. There is also precedent for efficiency recovering with Reynolds number in exactly this class of machine, as a NASA ten times scale test of a low specific speed rocket pump took the efficiency from 58.7% to 68.9% on Reynolds number alone [5].

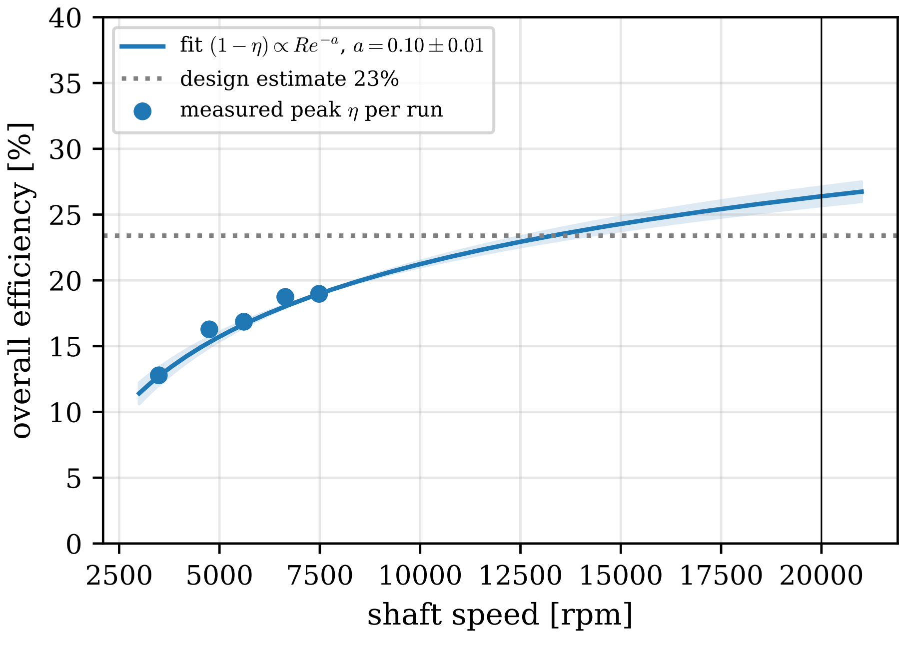


Figure 33: Peak overall efficiency per run against shaft speed, with the fitted loss ratio $(1-\eta) \propto Re^{-a}$ and its uncertainty band extrapolated to the design speed. The fitted exponent $a = 0.098 \pm 0.010$ matches the textbook friction scaling.

For context on whether around 19% measured and around 26% extrapolated is good or bad, the conventional counterfactual exists at literally this specific speed. Kim's closed impeller design at the same n_q reaches 67% in CFD, but it is unvalidated and requires 0.31 mm closed channels, the geometry this project's own conventional design attempt showed to be unmanufacturable [2]. The CFD studies of Barske type machines report higher efficiencies than measured here at higher Reynolds numbers [1, 3], while the historic measured partial emission pumps sit at 35 to 60% at much larger scale. The gap between this machine and those numbers is accounted for by the measured churning and the sub design Reynolds number, not by an unexplained deficiency.

One tension should be stated openly because it bounds the extrapolation. Carrying the churning multiplier of 2.5 unchanged to 20 000 rpm would predict around 3 kW of churning alone, more than the entire turbine power budget. The design efficiency therefore requires the multiplier to fall with Reynolds number, which is the same physics the fitted exponent already encodes. The multiplier is measured at test Reynolds numbers, and its Reynolds dependence is the explicit assumption of the extrapolation.

5.4 Pump: Cavitation

Specifically for rocket engine applications, the cavitation behaviour is of high importance. The following graph shows the cavitation curve generated by plotting the head at which cavitation was observed at different flow rates:

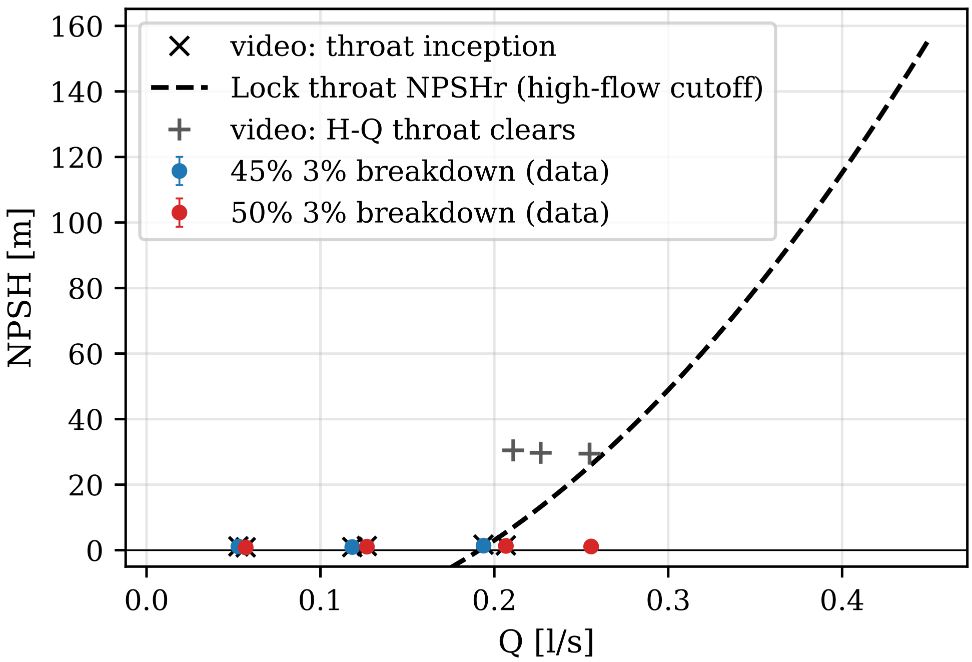


Figure 34: NPSH at 3% head breakdown against flow rate, with the visually observed throat cavitation onsets and the Lock throat cutoff curve. Error bars as in fig. 30.

fig. 34 shows the NPSH required to avoid cavitation at different flow rates. The dashed line is the theoretical curve generated by solving eq. (27) for the flow rate at which the head at the diffuser throat is equal to the vapor pressure head. Two interesting observations here.

The full suction test behaviour is shown in fig. 35 in the standard form, head normalised by the fully wetted reference against NPSH available, one curve per held flow rate, the same presentation Gülich uses for suction tests at constant speed. The curves hold their reference head over most of the suction range and then drop over a cliff. The 3% criterion line picks the breakdown point off each curve, and the cliff is steep enough that the choice of criterion barely moves the answer, as the head loses around 60% over a range comparable to the binning noise.

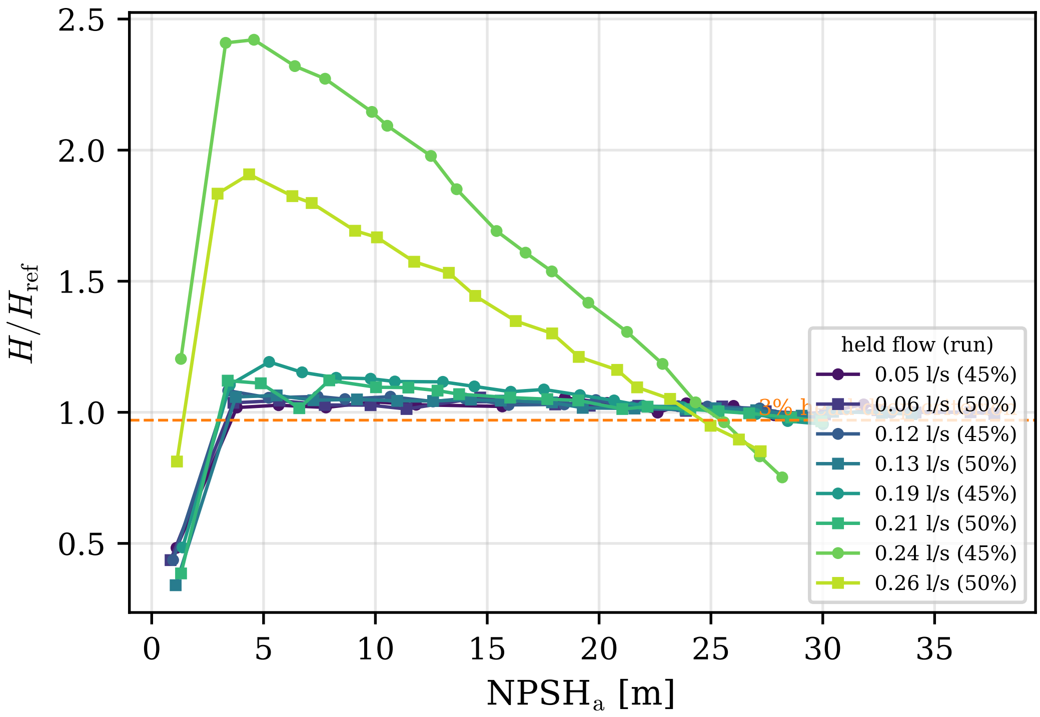


Figure 35: Suction test curves at constant speed: normalised head against NPSH available, one curve per held flow rate, by successive reduction of the inlet pressure. The horizontal line is the 3% head-drop criterion used for fig. 34.

The result that makes this section more than a standard suction test is where the breakdown happens. Two independent methods agree: the 3% breakdown from the pressure data coincides with the visually observed throat cavitation to around 0.1 m of NPSH, while cavitation at the impeller eye appears far earlier and does nothing measurable to the head. The naive reading of pump cavitation, bubbles at the inlet eye killing the head, is simply wrong for this machine. The eye cavitates benignly, and it is the diffuser throat that drives breakdown. This is Lock's diffuser limited claim, confirmed and this time directly visualised. One caveat applies to the measurement method: the inlet throttle drops the flow as well as the suction pressure, so the rigorous breakdown metric is the cliff knee rather than a fixed reference percentage, and the difference is small here because the cliff is steep.

Packaged as a single literature comparable number, the suction specific speed at breakdown is around 75 in metric form. That is close to the cavitation onset value of around 64 reported for a Barske type impeller in the same regime [1], and far below the industrial figure of around 200, because in this architecture the throat sets breakdown rather than the eye.

The throat sizing itself also lands in an interesting place against the literature. The throat to tip diameter ratio of this pump, about 0.051, follows the classical Barske and Balje guidance, the same starting point Danieli used for his partial emission rocket pump, whose initial throat ratio of 0.064 failed his head requirement. His CFD optimisation recovered head and efficiency by opening the throat ratio to 0.18 to 0.28 [3]. That independently corroborates the finding here that the classical throat under delivers and limits the runout, and it hands the design a cited future work item: enlarge the throat, traded against the suction margin that partial emission buys. His flow rate is 2.7 times larger and his study is CFD only, so it is the trend that is being cited, not the numbers.

(a) Fully wetted

(b) Eye cavitation only

(c) Throat cavitation

(d) Both sites, near breakdown

Figure 36: Stills through the transparent front casing at descending NPSH. The eye cavitates first with no measurable head effect, while the appearance of throat cavitation coincides with the head breakdown. Frame timestamps are synchronised to the logged data.

5.5 Coupled Turbopump

The integrated test is the proof of concept of the whole project. With the turbine driving the pump on shop air, the shaft settled at around 5400 RPM, the pump produced a measurable head rise, and the assembly survived repeated runs. The settling itself is a result: the speed approached the match point monotonically and held it without any active control, which is the self regulating behaviour predicted by the torque matching argument of section 2. The operating point is deep off design, with the stator pressure at roughly 38% of the design supply and a blade to jet speed ratio of around 0.06 against the design 0.2. fig. 37 shows the coupled pump data landing on the motor driven curves, which is the cross validation between the two test configurations.

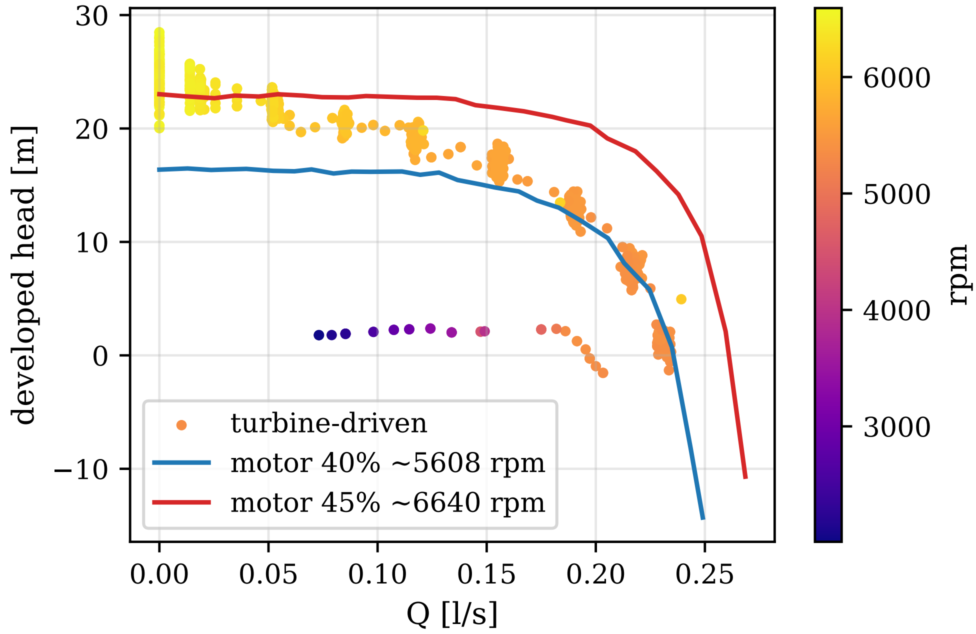


Figure 37: Coupled turbine driven pump operation against the motor driven characteristic curves. The coupled data, coloured by shaft speed, lands on the same head flow behaviour measured under motor drive.

The turbine side needs more care, because the rig has no air mass flow meter. Without a measured mass flow, a single forward efficiency number for the turbine would be under determined: a choked mass flow with a weak jet and a reduced mass flow with a near design jet both fit the same measured torque. Quoting one number would be dishonest, so this thesis does not.

The torque models compared against the data in fig. 38 are built from the Euler equation with the choked mass flow of eq. (63), which remains valid regardless of the rotor state. For a rotor receiving an effective tangential jet velocity c^e_{3u}, the shaft torque is

$$\tau = \dot{m}\,(1+\varphi_r)\,r_m\,(c^e_{3u} - u) \tag{90}$$

where the factor $(1+\varphi_r)$ carries the contribution of the turned relative flow at exit. The started model sets c^e_{3u} from the full design expansion at the measured supply pressure, while the unstarted model treats the effective jet as a fraction of it, fitted to the data.

What breaks the degeneracy is the starting analysis from section 3. At the test shaft speed the blade speed is so low that the rotor inlet relative Mach number rises to 1.58, above the Goldman starting limit of 1.408, so the rotor passages cannot swallow the startup shock system and the rotor runs unstarted. The design stage prediction was that the rotor cannot start below approximately 15 300 RPM, and the coupled test never exceeded half of that. An unstarted rotor sits behind a detached bow shock, the passage flow is subsonic, and the jet momentum reaching the blades collapses. fig. 38 shows the consequence: the measured torque of around 0.27 N m sits a factor of four below the started flow model at around 0.87 N m, and an unstarted model with an effective jet of around a third of the design velocity passes through the data. The near design jet branch of the degeneracy is ruled out precisely because it would require a started rotor, which the starting criterion forbids at this speed. The torque deficit is therefore not an anomaly, it is a diagnosed off design state that the design stage analysis predicted before the test was run.

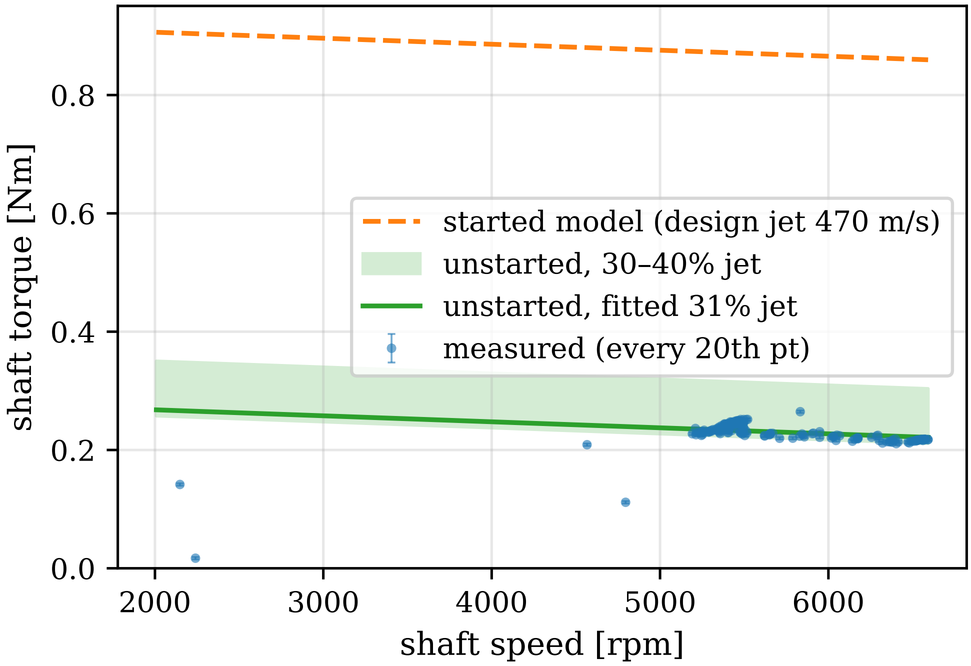


Figure 38: Coupled shaft torque against speed: measured data, the started flow model (dashed) and the unstarted model with a reduced effective jet (solid with band). The factor of four deficit against the started model is explained by the rotor running unstarted below the starting threshold.

Two rigour checks defend this diagnosis. First, the nozzle can be ruled out as the location of the loss. At the test supply pressure the over expanded conical nozzle sits at a pressure ratio of about 0.61 to ambient, above the separation threshold of roughly 0.4 from the rocket nozzle separation criteria [28], so the nozzle remains attached and supersonic and the velocity collapse must happen at the rotor. Second, the threshold itself should not be over read. With three discrete nozzles the rotor relative Mach number varies around the admitted arc, so the 15 300 RPM figure is a mean flow estimate and operation near it may start and unstart intermittently [9].
Because the stage was never measured in its started state, the design point efficiency prediction has to lean on external evidence, and the literature provides a tight bracket. The closest comparison is a machine of the same architecture, a radial inflow supersonic partial admission impulse stage at Mach 1.65, where the measured debit from ideal efficiency was 28 percentage points [21]. The Weiss chain here predicts 65% falling to 0.47, a debit of just under 28 points, so the loss model magnitude is corroborated experimentally to about one percentage point on the same machine class, with the honest boundary that his rotor ran started. Peng's 1.5 kW partial admission supersonic stage gives a CFD efficiency of around 42.5% [10], bracketing the prediction from below. NASA tested a Mach 2 stage from full admission down to 12.5% admission and found the full admission efficiency at 42.5% with the drop to 12.5% admission costing only around 5 points [29], which matters here because it acquits partial admission: at $\varepsilon = 0.15$ the admission penalty is modest, and the measured deficit belongs to the unstart, not the admission. Ohlsson's closed form partial admission theory, inside whose validation range this turbine's arc to chord ratio sits, composes with his low aspect ratio correlations to a literature only stage efficiency estimate of around 0.60 at his design speed ratio [14, 15], a second independent bracket. Finally, a precedent exists for exactly this failure mode: the SLA resin turbine Weiss cites also maxed out below its design speed with the cause left unexplained [24]. This work supplies the diagnosis that precedent lacked, the combination of supply droop and rotor unstart.

5.6 Extrapolation to the Design Point

The design point of 20 000 rpm was not reached in testing, so every claim about it is an extrapolation, and the value of this section is being precise about which extrapolations are licensed and which are not. On the pump side the licence is the non dimensional collapse of fig. 30. Because five speeds collapse onto one head coefficient curve, the affinity laws demonstrably hold for this machine, and fig. 39 carries the pooled curve to the design speed, where it passes through the design point. The efficiency extrapolation rides on the fitted Reynolds exponent of fig. 33 and arrives at 26.4% with a stated band.

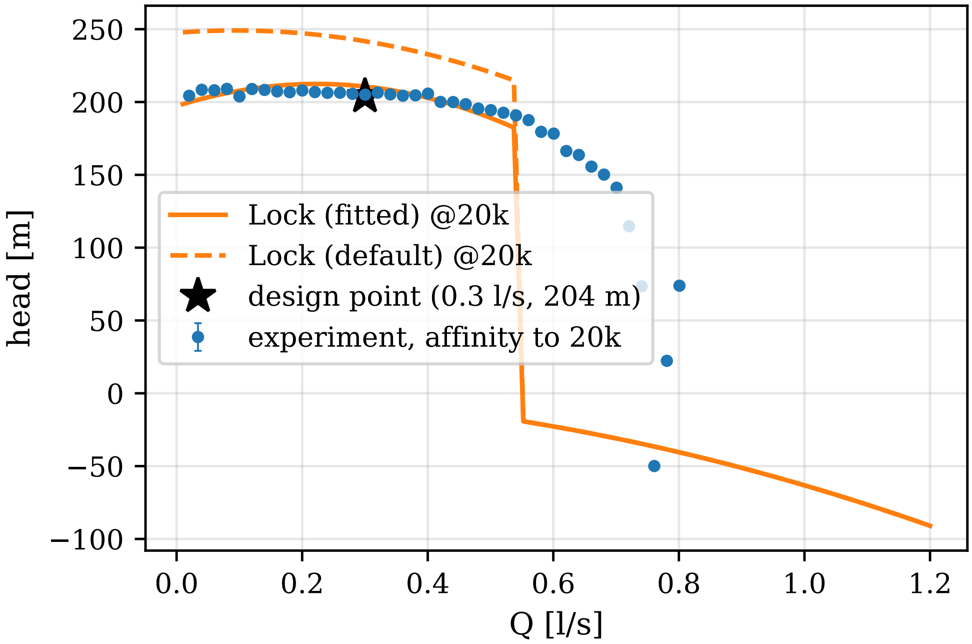


Figure 39: The pooled non dimensional characteristic carried to 20 000 rpm by the affinity laws, with the Lock model at design speed and the design point marked.

On the turbine side the honest statement is narrower. The coupled test ran at around 5400 RPM with the rotor unstarted, so the stage similitude parameters at design, the rotor relative Mach number and the blade to jet speed ratio, were never matched. What is validated are the speed independent pieces: the choked nozzle, the MoC profile generation against Goldman's published case, and the starting criterion itself, whose off design prediction the coupled test confirmed. Two named gaps remain on the road to 20,000 RPM. First, the started stage efficiency of 0.47 is a corroborated prediction, not a measurement, and stays that way until a test above the starting threshold. Second, the working fluid: cold air at $\gamma = 1.4$ stands in for hot gas at around 1.2, which shifts the expansion and the starting margin.

5.7 Synthesis

table 6 collects the quantitative comparisons of this chapter in one place.

Table 6: Predicted against measured summary. Percentage differences are computed by the analysis pipeline from the same registry that supplies every number in this thesis.

Quantity	Predicted	Measured	Δ	Note
head coefficient ψ	1.065	1.10 ± 0.05	+3 %	Barske design vs measured (5-speed colla
overall efficiency η at 20 000 rpm	23.4 %	27 %	+16 %	design estimate vs Re-fit extrapolation
shutoff churning power	Barske disk correlation	2.5 ×	+150 %	measured at shutoff, 15 points
rotor-inlet relative Mach M_{w3} at test	1.408	1.58	+12 %	vs Goldman starting limit → unstarted
coupled shaft torque	0.87 N m	0.27 N m	−69 %	started-flow model vs measured (unstart

On the adequacy of the toolchain, the verdict is positive within stated limits. Lock's model reproduces the head curve including its droop, and places the cavitation limited runout at the right flow once the as-built throat is used, so it is adequate for design stage screening of this machine class. The Barske sizing model delivered a pump that meets its design head coefficient within 3%. On the turbine, the speed independent codes validated cleanly, and the loss chain prediction is corroborated externally to about a point, but the started stage remains unverified by this rig. The 1D and analytical toolchain was sufficient to design, predict and diagnose everything this project measured, and nothing in the data demanded CFD.

On the Barske architecture itself, the central question of the thesis, the data supports a measured verdict. The architecture does what it promises at this specific speed: it is manufacturable where the conventional impeller is not, it delivers its design head, and its suction behaviour is dominated by a single identifiable mechanism that a transparent casing can watch. The price is efficiency, with churning measured at two and a half times the textbook estimate eating two thirds of the shaft power at test speed. Whether around a quarter efficiency at design speed is acceptable depends entirely on the application, and for expendable small rocket stages where feed pressure mass dominates, it plausibly is.

One mechanical learning point belongs in the record. The first rig iteration used printed plastic load bearing shafts, which crept and failed under high speed loads, and the redesign moved every rotating load path to metal, a stainless pump shaft and an aluminium turbine shaft. Printed structural parts are acceptable for casings and blading at these loads, but not for shafts.

6 Conclusions

A small scale low specific speed turbopump, a Barske pump driven by a supersonic partial admission impulse turbine, was designed, built and tested, and its measured behaviour was used to answer the four objectives set in section 1.4.

Objective 1, pump characterisation. The Barske pump delivers its design head, with the measured head coefficient of 1.10 ± 0.05 sitting 3% above the design value and collapsing across five speeds, which demonstrates dynamic similarity. The peak measured efficiency is $(19 \pm 2)\,\%$, and the dominant loss was identified and measured rather than assumed: churning of the flooded casing absorbs about two thirds of the shaft power at test speed, at 2.5 to 2.9 times the textbook disc friction estimate, measured directly at shutoff. The cavitation behaviour was characterised and, through the transparent casing, visualised: the impeller eye cavitates early and benignly, while breakdown is driven by the diffuser throat, with the 3% breakdown and the visual throat onset agreeing to around 0.1 m.

Objective 2, turbine design and model validation. The turbine was designed with a first principles toolchain of MoC blade generation, the Sasman and Cresci separation check, the Goldman starting criterion with Colclough's empirical derating, and the Weiss loss chain, giving a predicted total to static efficiency of 0.47 against an ideal 65%. The standalone brake test was not performed, so the started stage efficiency remains a prediction, corroborated to about one percentage point by a measured machine of the same architecture. The starting criterion proved to be the load bearing piece of the toolchain, as its off design prediction explained the coupled test result.

Objective 3, integration. Coupled turbopump operation was demonstrated at around 5400 RPM, with the turbine driving the pump to a measurable head rise and the shaft settling at a stable self regulating match point with no active control. The gap to the design speed is explained by two named and quantified mechanisms, the supply pressure droop to roughly 38% of the design stator pressure and the predicted rotor unstart below approximately 15 300 RPM, rather than by any unexplained deficiency of the machine.

Objective 4, toolchain assessment. The 1D and analytical toolchain is sufficient for design stage prediction of this class of machine. Lock's model reproduces the measured head droop and locates the cavitation limited runout once the as-built throat geometry is used, the affinity laws hold demonstrably, and the fitted Reynolds scaling carries the efficiency to 26.4% at the design speed with a stated band. No result in this work required CFD to predict or to explain.

Overall, this work provides an open, experimentally validated characterisation of a Barske turbopump at $n_q \approx 6.4$, including a measured loss breakdown and a direct visualisation of the cavitation mechanism that sets its operating limit.

6.1 Future Work

Each item below names the uncertainty it resolves.

1. Standalone turbine characterisation against the hysteresis brake, above the starting threshold of approximately 15 300 RPM. The configuration is built but was never run, and it resolves the single largest open prediction, the started stage efficiency of 0.47.

2. A 15 mm direct rigid coupler to unlock the full 20,000 RPM envelope of the torquemeter, removing the drive side speed limit on pump testing.

3. An enlarged diffuser throat, traded against suction margin. The measured throat limited runout and the published optimisation trend [3] both point at the same lever.

4. Hot gas generator products as the working fluid, closing the specific heat ratio gap between cold air at $\gamma = 1.4$ and combustion gas near 1.2, which also shifts the starting margin.

5. A machined metal impeller, to isolate the contribution of the printed surface quality to the measured churning multiplier.

6. A full admission variant at larger diameter, to isolate the partial admission sector losses from the rest of the loss budget.

7. CFD of the as-built geometry with the measured boundary conditions, now that there is experimental data to validate it against, targeting the partial admission windage and the shock boundary layer interaction the 1D tools cannot resolve.

8. Global shutter imaging with a synchronised strobe through the existing transparent casing. The present consumer camera resolves the cavity envelope but not bubble dynamics, since the frame rate sits far below blade passing and the rolling shutter distorts fast structures; individual bubbles already appear in isolated frames, so the optical access is proven and only the camera limits what it can measure.

9. High rate burst logging on the pressure channels. The logged data are 32 sample averages at 50 Hz, so blade passing unsteadiness at several hundred hertz is unresolvable by design; short bursts at the ADC's native rate would resolve it without changing the architecture.

References

[1] E. G. Knyazeva and Y. Klochkov. Design and fluid dynamics features of a low-flow high-head pump with barske-type impeller. E3S Web of Conferences, 320:04013, 2021.

[2] Hye In Kim, Tae-Seong Roh, Hwanil Huh, and Hyoung Jin Lee. Development of ultra-low specific speed centrifugal pumps design method for small liquid rocket engines. Aerospace, 9(9), 2022.

[3] Piero Danieli, Massimo Masi, Riccardo Tridello, Mario Pessana, Francesco Barato, and Alessandro Chiesa. Conceptual design of the electric pumps for the high-performance engine of a european lunar lander. In ASME Turbo Expo 2023: Turbomachinery Technical Conference and Exposition. American Society of Mechanical Engineers Digital Collection, 2023.

[4] OE BALJE. Turbomachines - A Guide To Design, Selection, and Theory. John Wiley & Sons, New York, 1981.

[5] C. Macgregor and A. Csomor. Rotating and positive-displacement pumps for low-thrust rocket engines. volume 1: Pump evaluation and design. Technical Report R-8494-1-VOL-1, NASA, 1974.

[6] Emanuel Boxer, James R. Sterrett, and John Wlodarski. Application of supersonic vortex-flow theory to the design of supersonic impulse compressor- or turbine-blade sections. Technical Report NACA RM L52B06, NACA, 1952.

[7] L. J. Goldman and V. J. Scullin. Analytical investigation of supersonic turbomachinery blading. 1 - computer program for blading design. Technical Report NASA-TN-D-4421, NASA, 1968.

[8] L. ERIKSSON. Development of a supersonic turbine stage for the hm60 engine. In 20th Joint Propulsion Conference. American Institute of Aeronautics and Astronautics.

[9] Samuel Sudhof. Development Techniques for Supersonic Turbines. PhD thesis, Technical University of Munich, 2020.

[10] Ningjian Peng, Enhua Wang, and Wenli Wang. Design and analysis of a 1.5 kw single-stage partial-admission impulse turbine for low-grade energy utilization. Energy, 268:126631, 2023.

[11] Ahmed M. K. Abdalhamid and Ahmed Eltaweel. Design and analysis of a single-stage supersonic turbine with partial admission. Energy, 309:133100, 2024.

[12] Johann Friedrich Gülich. Centrifugal Pumps. Springer International Publishing, Cham, 2020.

[13] Thomas Lock. A forced vortex pump for high speed, high pressure, low flow applications. SAE Transactions, 74:729–737, 1966.

[14] Gunnar O. Ohlsson. Partial admission turbines. Journal of the Aerospace Sciences, 29(9):1017–1024, 1962. PDF in Downloads (ohlsson1962.pdf) – import to Zotero, then move this entry to references.bib.

[15] Gunnar O. Ohlsson. Low aspect ratio turbines. Journal of Engineering for Power, 86(1):13–16, 1964. In Zotero (storage JJQF4B5G) – citekey guessed per BBT scheme; VERIFY against the final BBT export.

[16] Arthur J. Glassman. Turbine design and application volumes 1, 2, and 3. Technical Report NASA SP-290, NASA, 1994.

[17] L. J. Goldman. Analytical investigation of blade efficiency for two-dimensional supersonic turbine rotor blade sections. Technical Report NASA-TM-X-2095, NASA, 1970.

[18] C. D. Colclough. Design of turbine blades suitable for supersonic relative inlet velocities and the investigation of their performance in cascades: Part i-theory and design. Journal of Mechanical Engineering Science, 8(1):110–128, 1966.

[19] SKF Group. Rolling Bearings. AB SKF, 2018. Friction model: rolling, sliding and drag moment components. Verify catalogue edition used for the constants in bearing.py.

[20] Liquid rocket engine turbines. Technical Report NASA SP-8110, NASA, 1974.

[21] Liam Donnelly. The design and experimental verification of a compact supersonic centripetal impulse turbine. Meng final year project, Imperial College London, Department of Aeronautics, 2025.

[22] S. Kawasaki, T. Shimura, M. Uchiumi, M. Hayashi, and J. Matsui. Unsteady response of flow system around balance piston in a rocket pump. In Progress in Propulsion Physics, pages 457–466, St. Petersburg, Russian, 2013. EDP Sciences.

[23] John D. Anderson. Modern Compressible Flow: With Historical Perspective. McGraw-Hill, 3 edition, 2003. Turbulent recovery factor r = Pr^(1/3). Verify edition/page before final.

[24] Andreas P. Weiß, Václav Novotný, Tobias Popp, Philipp Streit, Jan Špale, Gerd Zinn, and Michal Kolovratník. Customized orc micro turbo-expanders - from 1d design to modular construction kit and prospects of additive manufacturing. Energy, 209:118407, 2020.

[25] Wilfred E. Baker, P. A. Cox, P. S. Westine, J. J. Kulesz, and R. A. Strehlow. Explosion Hazards and Evaluation. Elsevier, 1983.

[26] Paul J. Hazell. Armour: Materials, Theory, and Design. CRC Press, 2015.

[27] Muhammad Fasahat Khan. Design and optimization of low-specific speed centrifugal pump.

[28] Ralf H. Stark. Flow separation in rocket nozzles, a simple criteria. Technical Report AIAA 2005-3940, 41st AIAA/ASME/SAE/ASEE Joint Propulsion Conference, Tucson, AZ, 2005.

[29] Thomas P. Moffitt and Jr. Klag, Frederick W. Experimental investigation of the partial-admission performance characteristics of a single-stage mach 2 supersonic turbine. Technical Report NASA TM X-80, NASA Lewis Research Center, 1959.

[30] American Petroleum Institute. API standard 521: Pressure-relieving and depressuring systems. American Petroleum Institute, 2014.

A Conventional Impeller Design Attempt

B Supplementary Safety Calculations

The containment and overspeed calculations that sized the protection hardware are in the main body, eq. (80) onwards. This appendix carries the remaining two analyses of the safety case: the venting noise and the venting reaction force. As in the body, the numbers are the 100 bar worst case for the stored air system that was later descoped to 7 bar shop air, retained because the hardware was retained.

B.1 Venting Noise

The sound level of gas venting is estimated per API 521 [30]. The choked mass flow through the vent restriction is

$$\dot{m} = A_{choke}\, P_0\, \frac{\Gamma}{\sqrt{T}}, \qquad \Gamma = \sqrt{\frac{\gamma}{R}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \tag{91}$$

and the sound pressure level at 30 m and at distance r follow from

$$L = 5\log_{10}\left(\frac{P_0}{P_e}\right) + 51.5, \qquad L_{30} = L + 10\log_{10}\left(\tfrac{1}{2}\dot{m}c^2\right), \qquad L_p = L_{30} - 20\log_{10}\left(\frac{r}{30}\right) \tag{92}$$

For the worst case 100 bar vent through a 1/8” line this gives 102 dB at 30 m and 131 dB at 1 m. Against the Control of Noise at Work Regulations 2005 limits, nobody is in the room during venting, and the 27 dB SNR ear defenders worn in the control room bring the exposure below the action level.

B.2 Venting Reaction Force

The reaction force of an open vent is

$$F = \dot{m}\, c + A\left(P_{choked} - P_e\right) \tag{93}$$

with c the speed of sound at the choked condition. At 100 bar a 1/2” vent would produce around 1.6 kN, which is why the vent line is reduced to 1/8”, bringing the force to around 100 N against a 50 kg weighted stand, with all vents pointed upward so the reaction acts into the floor.

C Additional Figures and Tables